PhyDiff U-Net: A Physics-Informed Diffusion-Inspired Transformer for Infrared Small-Target Segmentation

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ABSTRACT

Infrared small-target segmentation (IRSTS) task is challenging because the far away targets are often only a few pixels, which only provide weak structural information. Complex background clutter, such as clouds, terrain edges, and reflections, can further make these targets difficult to distinguish. Existing CNN- and Transformer-based methods mainly rely on discriminative feature learning, but repeated downsampling and feature transformation can weaken small-target responses. Moreover, relatively few approaches jointly incorporate physics-aware priors, feature restoration, and uncertainty modeling within a unified framework. To address these shortcomings, we propose a light-weight physics-informed diffusion transformer for Infrared small-target segmentation (IRSTS), named PhyDiff U-Net. The model uses both thermal-contrast and frequency priors to enhance weak target representations during encoding. At the bottleneck, an edge aware efficient Transformer captures global contextual relationships, while a noise-perturbation-guided feature refinement module restores weakened target features. During decoding, wavelet-based multi-scale reconstruction, spatial attention, and Sobel attention (SDA) module are employed to recover fine spatial details and target boundaries. Finally, a heteroscedastic uncertainty head estimates pixel-wise confidence to improve robustness in ambiguous regions. We test PhyDiff U-Net on three common benchmark datasets, namely NUAA-SIRST, IRSTS-1K and NUDT-SIRST, and the experimental results demonstrate the high robustness of PhyDiff U-Net under different conditions. The model achieves 76.87%, 65.75%, and 91.20% IoU scores respectively, with consistently high detection probability and low false-alarm rates. PhyDiff U-Net is light and efficient, with only 3.16M parameters and 5.732 GFLOPs, achieving a strong detection performance and reaching to 151.39 FPS. The integration of thermal and frequency priors, boundary guidance, seperable attention based noise-perturbation-guided feature refinement, and uncertainty estimation provides an efficient framework for robust Infrared small-target segmentation (IRSTS).

1. Introduction

1.1. Background and Motivation

Infrared small target segmentation (IRSTS) is an important task in computer vision with applications in defense, aerospace, environmental monitoring and automated sensing. Infrared small target segmentation and segmentation are supporting to a wide range of practical uses such as maritime surveillance, missile and early-threat detection, search and rescue, forest-fire monitoring, border security, and satellite surveillance.

Unlike visible-spectrum imaging, infrared imaging captures thermal radiation emitted by objects rather than relying primarily on external illumination. This allows infrared sensors to operate during both daytime and nighttime and makes them relatively less dependent on visible illumination conditions. However, the advantages of infrared sensing are accompanied by fundamental challenges when the sensing distance becomes large. As the distance between the sensor and the target increases, the target may occupy only a few pixels in the image. At the same time, sensor noise, thermal diffusion, atmospheric turbulence, optical blur, and background radiation can significantly degrade the target response.

A small infrared target is defined as an object that occupies an area of no more than $\sim 9 \times 9$ pixels, or less than 0.15% of the image area, following the SPIE convention Liu et al. (2025). Such targets contain very limited structural information and may lack distinctive shape, texture, and intensity-gradient characteristics. Consequently, they frequently appear as weak local intensity variations embedded within complex backgrounds, including clouds, mountainous terrain, sea waves, reflections, and building edges.

The problem becomes especially hard when the contrast between target and background is low. In many infrared scenes the signal-to-clutter ratio (SCR) may be of the order of 1–3, so that the target response is comparable to background fluctuations. Furthermore, background structures can exhibit intensity patterns that resemble genuine

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targets, resulting in false alarms. The combination of extremely small spatial extent, weak thermal contrast, limited structural information, and complex background clutter therefore makes IRSTS substantially more difficult than conventional semantic segmentation.

1.2. Limitations of Existing Work

Inspite of the substantial progress in Infrared small-target segmentation (IRSTS), there are still some limitations.

First, most existing deep learning methods do not explicitly take into account the physical characteristics of the infrared image formation process. Infrared observations are affected by thermal radiation, atmospheric transmission, optical point-spread functions (PSFs), focal plane sampling and electronic noise of the sensor. Nevertheless, many learning-based methods treat the input primarily as a grayscale image and rely on data-driven feature discrimination. Ignoring these physical factors can reduce robustness when the imaging conditions or background characteristics change.

Second, small targets are highly vulnerable to irreversible information loss during feature extraction. Conventional convolutional and Transformer-based segmentation networks frequently employ multiple downsampling operations, with effective spatial reductions of approximately 4–32 times in different stages Ronneberger et al. (2015); Chen et al. (2018). If the target initially occupies only a couple of pixels, repeated downsampling can strongly attenuate its representation or even completely remove it. Thus, in the following feature reconstruction it is necessary to recover information which may have been already heavily degraded.

Third, uncertainty is rarely considered explicitly in deterministic segmentation frameworks. Most existing networks generate a binary or probabilistic segmentation mask without providing a calibrated estimate of confidence. In ambiguous regions containing weak targets or target-like clutter, a model may produce a confident but incorrect prediction. For operational applications, particularly surveillance and early-warning systems, understanding the reliability of individual predictions is therefore important.

1.3. Key Insight

The limitations motivate a different perspective, where instead of directly segmenting a severely degraded infrared observation, the network should first restore physically plausible target information and then perform segmentation with explicit confidence estimation.

The main idea of this work is to combine physics-conditioned feature restoration and uncertainty-aware segmentation. Thermal contrast and frequency domain information provide a complementary description of the target saliency Infrared imaging priors limit the feature restoration process. A noise-perturbation-guided feature refinement mechanism is then used to restore weakened target representations before the final segmentation stage. Rather than relying exclusively on discriminative feature learning, the proposed framework therefore attempts to preserve or reconstruct target information that may have been weakened during imaging and feature extraction.

A Transformer-based feature encoder is further employed to model long-range contextual relationships while retaining small-target responses. Finally, an uncertainty estimation module provides pixel-level confidence information, allowing the network to distinguish reliable predictions from ambiguous regions.

This combination of physical modeling, generative feature restoration, discriminative segmentation, and uncertainty estimation forms the central design principle of PhyDiff U-Net.

1.4. Contributions

We summarize the main contributions of this paper as follows:

- We propose a physics-informed prior extraction mechanism by combining the thermal-contrast information and high-frequency saliency to provide physically meaningful guidance for infrared small target representation learning .
- We propose a Feature Restoration Enhancement module at the bottleneck of the network to restore weakened small-target features prior to segmentation. The refinement process takes advantage of the prior consistency to improve the target responses while suppressing the background clutter.
- We employ an edge aware efficient Transformer for bottleneck feature processing. Separable self-attention is employed to capture long-range contextual dependencies with approximately linear computational complexity, while an adaptive edge-aware mechanism enhances weak and ambiguous target boundaries.

- We design a boundary-aware decoding and uncertainty estimation framework that combines multi-scale reconstruction, spatial attention, and boundary guidance with pixel-level uncertainty estimation to improve localization and reduce false alarms.
- We perform extensive experiments on three public infrared small target datasets, i.e., NUAA-SIRST, IRSTS-1K, and NUDT-SIRST, and evaluate the segmentation performance, detection probability, false-alarm behavior, uncertainty calibration, computational efficiency, ablation experiments, and failure-case analysis.

1.5. Paper Organization

The rest of this paper is organized as follows. Section 2 reviews related work in model-driven Data-driven segmentation, infrared small target segmentation, generative models based on diffusion, physics-informed learning and Transformer architectures for small objects. The proposed PhyDiff U-Net framework including the physics-prior extraction mechanism, Feature Restoration Enhancement (FRE), boundary guided efficient Transformer, multi-scale decoder and uncertainty estimation is presented in Section 3. The datasets, implementation details, evaluation metrics and experimental protocols are described in Section 4. Section 5 describes experimental results, comparison, ablation studies, efficiency analysis, and failure cases. Finally, Section 6 concludes the paper and outlines possible future directions.

2. Related Work

2.1. Model-Driven Infrared small target segmentation

Model-driven Infrared small target segmentation methods are primarily based on manually designed mathematical models and prior assumptions. Small target has less structural information and traditional methods usually try to enhance the responses of local targets with the suppression of background variations in the surrounding background.

Morphological techniques are based on erosion and dilation operations, for example, the Top-Hat transform, and are used to identify bright structures that differ from their local surroundings Wang and Wang (2021). However, their performance may be seriously degraded in complex infrared scenes. Top-Hat based approaches have been reported to achieve nIoU values below 15% and false-alarm rates larger than $300 \times x10^{-6}$ on the IRSTS-1K dataset Zhang et al. (2022a).

Sliding-window approaches, including Max-Mean and Max-Median filtering, exploit local statistical differences between candidate targets and their neighborhoods Kim et al. (2009); Deshpande et al. (1999). These methods generally depend on assumptions concerning target geometry, local contrast, or signal-to-noise characteristics and can therefore become unreliable when the background contains target-like structures.

There has also been extensive study of saliency methods inspired by the human visual system. Local Contrast Measure (LCM), Multiscale Patch-based Contrast Measure (MPCM), Weighted Local Difference Measure (WLDM), Tri-Layer Local Contrast Measure (TLLCM) are representative ones Wang et al. (2023); Xi et al. (2025); Zheng et al. (2023); Kou et al. (2022). While such methods can enhance local contrast, they are not very effective if the target has low contrast or if the clutter has similar local patterns. Related methods have been reported to have detection probabilities lower than 50% on IRSTS-1K with relatively high false-alarm rates Zhang et al. (2022a).

Low-rank and sparse decomposition methods model infrared image as superposition of structured background and sparse target. The infrared patch-image (IPI) model and its variants formulate target detection as a low-rank and sparse recovery problem Lu et al. (2022). Later techniques such as RIPT, PSTNN and multi-frame tensor based methods extend this idea to better suppress background and separate target Su et al. (2023); Yan and Guo (2024); Zhu and Feng (2026).

However, these methods remain sensitive to violations of their underlying assumptions. For example, low-rank background modeling can suppress genuine target information when target and background structures become highly correlated. On IRSTS-1K, IPI, RIPT, and PSTNN have been reported to obtain nIoU values of 13.51%, 11.43%, and 13.71%, respectively, with corresponding limitations in detection probability and false-alarm suppression Liu et al. (2024a). Complex clutter can further violate the assumed low-rank structure Liang et al. (2022). In addition, iterative optimization procedures can impose considerable computational costs, limiting their suitability for real-time applications.

2.2. Data-Driven Infrared small target segmentation

Infrared small-target segmentation (IRSTS) has greatly benefited from advances in deep learning which enables models to learn discriminative representations directly from data. In the experiments, CNN based architectures have shown much better performance than traditional hand-crafted methods on widely used IRSTS benchmarks.

The Asymmetric Context Module (ACM) achieved 69.18% nIoU on NUAA-SIRST and 34.03% nIoU on IRSTS-1K Dai et al. (2021a); Zhang et al. (2022a). Later, ALCNet modeling local contrast improved the performance to nIoU of 71.05% and 67.20% on NUAA-SIRST and NUDT-SIRST respectively Dai et al. (2021b).

DNANet proposed the dense nested attention mechanism and achieved nIoU of 79.20%, 88.58% and 66.38% on NUAA-SIRST, NUDT-SIRST and IRSTS-1K respectively Li et al. (2023a). It has strong segmentation ability, but its relatively complex architecture and purely data-driven learning leave room for improvement in physical robustness Liu et al. (2026).

Other methods such as AGPCNet and RDIAN have investigated mechanisms of contextual and reverse-information for small target representation learning Zhang et al. (2023); Sun et al. (2023). RDIAN reached nIoU of 75.39%, 79.14%, and 59.72% on NUAA-SIRST, NUDT-SIRST, and IRSTS-1K respectively Zhang and Yin (2025); Sun et al. (2023).

ISNet and UIUNet Zhang et al. (2022a); Wu et al. (2023b) proposed more efficient feature interaction and reconstruction strategies. UIUNet achieved nIoU 93.89% on NUDT-SIRST, 79.99% on NUAA-SIRST, and 66.66% on IRSTS-1K. However, more sophisticated feature aggregation mechanisms may increase the complexity of the architecture.

Some recent architectures such as MTU-Net, SCTransNet and MSHNet have explored multiscale feature extraction, Transformer-based interaction and high-resolution target representation Wu et al. (2023a); Yuan et al. (2024); Liu et al. (2024a). SCTransNet achieves 81.08%, 94.38% and 68.15% of nIoU on NUAA-SIRST, NUDT-SIRST and IRSTS-1K, respectively. In the reported evaluation, MSHNet achieved competitive performance with nIoU of 38.29% and false-alarm-rate of $16.47 \times x10^{-6}$.

Despite these improvements, most data-driven methods still process infrared images primarily as grayscale observations. Consequently, the underlying thermal radiation process, optical blur, atmospheric effects, sensor noise, and sampling mechanisms are not explicitly represented. In addition, repeated feature downsampling can weaken extremely small target responses, while deterministic segmentation outputs generally provide limited information about prediction reliability.

These limitations motivate the integration of physical imaging priors, generative feature restoration, and uncertainty estimation into a unified segmentation framework.

2.3. Diffusion Models and Generative Super-Resolution

Diffusion models have gained popularity as powerful generative frameworks for learning complex data distributions . Denoising diffusion probabilistic models (DDPMs) describe generation as a gradual forward degradation process, and then a learned reverse denoising process Ho et al. (2020). Score-based stochastic differential equation methods offer a continuous-time formulation of similar forward and reverse processes Song et al. (2021).

Conditional diffusion models have also shown good performance on image restoration and super-resolution problems. SR3 uses conditional diffusion to restore high-resolution image content from low-resolution observations step by step Saharia et al. (2023). Diffusion-based approaches have been studied for segmentation and object detection Liu et al. (2024b).

However, directly applying conventional diffusion models to Infrared small-target segmentation (IRSTS) presents two major challenges. First, generative models can potentially hallucinate structures that do not correspond to genuine targets. In infrared surveillance, such hallucinations can translate directly into false alarms. Second, conventional diffusion inference can require hundreds or even thousands of iterative denoising steps, creating a substantial computational burden and making real-time deployment difficult.

Hence, an effective small target framework in infrared must leverage the restoration capability of diffusion models while enforcing the process with physically meaningful information. To this end, PhyDiff U-Net proposes physics-conditioned feature refinement and prior-consistency constraints, and an efficient inference strategy instead of a full high-step generative sampling procedure .

2.4. Physics-Informed Deep Learning

Physics-informed deep learning offers a mechanism to integrate domain knowledge and physical constraints in data-driven models. For example, physics-informed neural networks (PINNs) integrate governing physical equations or differential constraints into the optimization process Fan et al. (2025). More generally, physics-guided regularization can be used to restrict learned representations by known physical relationships Filipović et al. (2025).

For infrared imaging, incorporating the image formation process is particularly important because observed pixel intensities depend on thermal radiation, atmospheric transmission, optical point-spread functions, focal-plane sampling, and sensor noise.

NUDTNet introduced optical PSF information into Infrared small target segmentation through Gaussian-matched adaptive filtering Wu et al. (2025). Other PSF-aware networks have similarly attempted to account for optical degradation during feature learning Zhao et al. (2018). Atmospheric and radiometric modeling has also been investigated by considering transmittance and path-radiance effects in infrared observations Wu et al. (2021).

PI-SAM incorporates physical properties, including local brightness contrast, signal-to-noise ratios, and center-surround differences, into Infrared small-target segmentation (IRSTS) learning Li et al. (2026). PFDMNet utilizes fractional Fourier transform and wavelet-related representations to exploit frequency-domain information Hu et al. (2026). DPNet integrates anisotropic diffusion and thermal concentration characteristics into target modeling Tong et al. (2026).

These studies demonstrate that physical information can improve infrared target representation. Nevertheless, physics-aware mechanisms are often introduced as individual components rather than as part of a unified framework combining physical priors, generative feature restoration, Transformer-based global modeling, boundary-aware decoding, and uncertainty estimation.

2.5. Transformers for Tiny Objects

Transformers have shown remarkable ability to model long-range dependencies via self-attention . Vision transformer (ViT) and hierarchical architectures like swin transformer have been employed in many visual recognition and segmentation tasks, following the successful application of the Transformer architecture in natural language processing.

However, tokenization introduces a particular challenge for infrared small targets. ViT-style architectures partition an image into patches before constructing visual tokens, which can disrupt the representation of objects occupying only a few pixels Pi et al. (2025). Although Swin Transformer reduces the computational cost through local shifted windows, the use of fixed spatial windows can still weaken extremely small target responses.

Thus, multi-scale target information preservation is very important for Infrared small-target segmentation(IRSTS). Some architectures, e.g., SCTransNet Yuan et al. (2024) and IST-TransNet Li et al. (2023b) have explored the Transformer based feature interaction and attention aggregation to improve the small target representation.

The proposed framework instead introduces an efficient edge-aware Transformer at the bottleneck, where high-level contextual relationships can be captured while the preceding physics-guided modules preserve weak target responses.

2.6. Research Gap

The reviewed literature reveals several complementary but unresolved limitations. Model-driven approaches explicitly exploit physical or statistical assumptions but are often inflexible in complex scenes. Deep learning approaches provide stronger discriminative representations but generally do not model the infrared imaging process explicitly. Diffusion models offer powerful restoration and generative capabilities but can introduce hallucinated structures and incur substantial computational costs. Transformer architectures improve contextual reasoning but may lose extremely small targets through tokenization and spatial reduction. Finally, most existing approaches provide deterministic segmentation masks without explicit uncertainty estimation.

As far as we know, there are few existing IRSTS methods that integrate thermal and frequency-domain physics priors, physics-guided generative feature restoration before segmentation, physical constraints in a noise-perturbation-guided feature refinement process, efficient Transformer-based contextual modeling, boundary-aware reconstruction and uncertainty estimation into a single framework.

PhyDiff U-Net explores this hybrid direction by integrating these complementary components into a unified architecture designed specifically for weak, small, and ambiguous infrared targets.

3. Proposed Method

3.1. Overall Framework and Problem Formulation

The input of our model is an infrared grayscale image $X \in \mathbb{R}^{1\times H\times W}$, where H and W are the height and width of the image, respectively, as shown in Fig. 1 and a three-layer encoding stage, $L_i, i \in \{1, 2, 3\}$. The encoder feature maps obtained are $\mathbf{F}_i^E \in \mathbb{R}^{C_i\times H_i\times W_i}$, where C_i, H_i and W_i are the number of channels, height and width of the feature map in the i-th encoding stage. Each encoding layer has a Physics Prior Extraction (PPE) module, since infrared small targets often have comparatively higher intensity than the surrounding background and create stronger local high-frequency responses due to the contrast between the target and neighbors. The PPE module exploits these properties to extract local thermal and frequency information to enhance potential target responses and suppress background regions with less information. The extracted physics-aware features are then fused with the original feature representation through element-wise addition or channel-wise concatenation. The encoder-decoder structure is mainly inspired from the original U-Net Ronneberger et al. (2015). However, instead of the conventional max pooling we use Discrete Wavelet Transform (DWT) Liu et al. (2019) for spatial downsampling. This problem becomes especially serious for infrared small-target detection, because the target can be covered by only a few pixels, and the target response may be degraded or even deleted by multiple pooling operations.

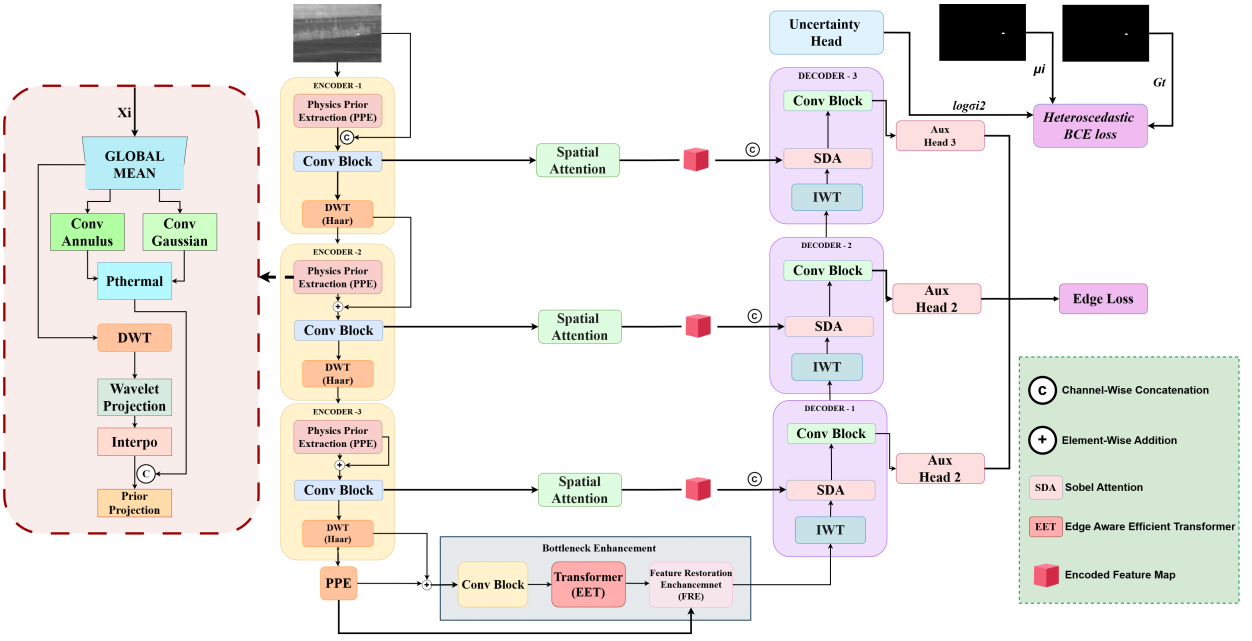


Figure 1: Proposed PhyDiff U-Net: overall architecture. The network is composed of three PPE-enhanced encoder stages based on Haar DWT downsampling, a bottleneck with Edge Aware Efficient Transformer (EET) and Feature Restoration Enhancement (FRE) and three decoder stages with IWT upsampling, spatial-attention-guided skip connections, and Sobel Attention (SDA). We make use of auxiliary prediction heads for deep supervision, and the uncertainty head estimates mean and log-variance for heteroscedastic BCE optimization. The enlarged panel on the left shows the internal structure of the Physics Prior Extraction (PPE) module.

CNN-based networks have been widely used in infrared small-target segmentation (IRSTS) and they are good at local feature extraction. However, their limited receptive field can restrict the ability to capture long range contextual information, especially when the targets vary significantly in size. Although atrous convolution **Chen et al. (2018)** can be used to solve this limitation, fine spatial information needs to be preserved as infrared small targets may be only a few pixels. In an effort to address this problem, Vision Transformers with self-attention **Dosovitskiy et al. (2021)** have been paid more and more attention, because of their capability to model global relationships between distant regions. Therefore, after extracting semantic features in the encoder, our **PhyDiff U-Net** adopts a transformer-based bottleneck to capture the global relationship between potential target regions and background regions. However, transformers are generally less effective in preserving fine local details, which are important for accurately reconstructing the

ambiguous edges of small infrared targets. To deal with this limitation, we use a **edge detection module Cui et al. (2021)** to guide the transformer to focus more on the patches with the boundary and generate a patch-wise attention map.Since the standard self-attention has a quadratic computational complexity in terms of the number of patches, we replace it with **separable self-attention Mehta and Rastegari (2023)** which is more computationally efficient. Finally, to mitigate the possible weakening of very small targets responses after several times of downsampling and feature transformation, we add controlled Gaussian perturbations to the bottleneck features during training. The previously extracted physics-prior features are then used to guide feature refinement, helping the network preserve and enhance weak target responses before decoding.

The decoding stage consists of three upsampling layers followed by an uncertainty head for pixel-wise target prediction in the network. We use the **Inverse Wavelet Transform (IWT) Liu et al. (2019)** to recover the spatial resolution and combine the information of the wavelet sub-bands. A 1×1 convolution is subsequently used to reduce the number of feature channels. Rather than directly concatenating the raw encoder features with the decoder features, we apply **spatial attention Woo et al. (2018)** to the encoder feature map $\mathbf{F}_i^E \in \mathbb{R}^{C\times H\times W}$, allowing the network to emphasize target-related regions before feature fusion. The fused feature map $\mathbf{F}_i^F$ is then convolved and fed into a **Sobel Attention (SDA) module** to enhance the target boundary information. Finally, the uncertainty head predicts both a target logit map $\boldsymbol{\mu}_i$ and a pixel-wise uncertainty map $\log \boldsymbol{\sigma}_i^2$, representing the segmentation response and its associated aleatoric uncertainty, respectively. The logits are converted to the final probability map $M = \sigma(\boldsymbol{\mu})$, where pixels with $M(i, j) > 0.5$ are classified as targets and the rest as background. Deep supervision Lee et al. (2015) in the decoder further assists the network in recovering spatial details and improving localization.

3.2. Physics Prior Extraction

3.2.1. Thermal Contrast Prior

Relying solely on predefined window or receptive-field sizes may be insufficient to effectively capture targets with different variations in scale. Therefore, instead of considering only the spatial size of the target, we further exploit two useful characteristics of infrared small targets, **local thermal contrast** and **high-frequency information**. Generally, an infrared small target has a higher intensity than its surrounding background, and has localized high frequency responses due to its small and compact structure. Given an input image $\boldsymbol{X}$, we first aggregate the information across the channel dimension to obtain a single-channel intensity representation:

$$I(i, j) = \frac{1}{C} \sum_{c=1}^{C} X_c(i, j). \tag{1}$$

where $\boldsymbol{C}$ is the number of channels and $\boldsymbol{I(i,j)}$ is the mean response at the spatial location $\boldsymbol{(i,j)}$.

To characterize the local thermal contrast of a potential target, we compare its Gaussian point-spread-function (PSF) response with the average intensity of its surrounding neighborhood. The thermal prior is defined as

$$I_{\mathrm{PSF}}(i, j) = (G_\sigma * I)(i, j), \tag{2}$$

$$I_{\mathrm{ann}}(i, j) = (A * I)(i, j). \tag{3}$$

where $\boldsymbol{G}$ denotes a Gaussian kernel and $\boldsymbol{A}$ represents an annulus-shaped kernel where, denotes the convolution operation.

Thus, the thermal contrast can be expressed as

$$P_{\mathrm{thermal}}(i, j) = I_{\mathrm{PSF}}(i, j) - I_{\mathrm{ann}}(i, j). \tag{4}$$

A large positive value of $\boldsymbol{P\mathrm{thermal}(i,j)}$ indicates that the central region is considerably brighter than its surroundings and is therefore more likely to correspond to an infrared small target.

3.2.2. Frequency Prior

In addition to thermal contrast, infrared small targets often produce localized high-frequency responses. To capture this complementary information, we employ the Discrete Wavelet Transform (DWT) Liu et al. (2019). The corresponding filters can be represented as

$$H_{LL} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \qquad H_{LH} = \frac{1}{2}\begin{bmatrix}-1 & -1\\ 1 & 1\end{bmatrix},$$
$$H_{HL} = \frac{1}{2}\begin{bmatrix}-1 & 1\\ -1 & 1\end{bmatrix}, \qquad H_{HH} = \frac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}. \tag{5}$$

The ***LL*** component mainly preserves the low-frequency structural information of the input, whereas ***LH, HL***, and ***HH*** capture directional variations and fine spatial details.

The wavelet decomposition of an input feature map x can be expressed as

$$F^i = \downarrow_2 \left(x \otimes H_i\right), \qquad i \in \{LL, LH, HL, HH\}. \tag{6}$$

where $\otimes$ denotes convolution and $\downarrow_2$ represents spatial downsampling by a factor of two.

The four wavelet sub-bands are subsequently combined using a learnable 1×1 convolution. At each spatial location, the resulting frequency representation can be written as

$$\mathbf{F}_i^W = \mathrm{Conv}_{1\times1}\left(\left[\mathbf{F}_i^{LL}, \mathbf{F}_i^{LH}, \mathbf{F}_i^{HL}, \mathbf{F}_i^{HH}\right]\right), \tag{7}$$

where $\mathbf{F}_i^{LL}$, $\mathbf{F}_i^{LH}$, $\mathbf{F}_i^{HL}$ and $\mathbf{F}_i^{HH}$ are the low-low, low-high, high-low and high-high frequency sub-bands respectively. A learnable 1x1 convolution maps the concatenated wavelet features to a compact representation while preserving spatial information from different frequency components.

$$F(i,j) = w_1 F^{LL}(i,j) + w_2 F^{LH}(i,j) + w_3 F^{HL}(i,j) + w_4 F^{HH}(i,j) + b. \tag{8}$$

where ***w1,w2,w3,w4*** and ***b*** are learnable parameters.

3.2.3. Physics-Prior Fusion

Since the ***DWT*** reduces the spatial resolution, the resulting frequency feature ***F*** is resized to the original spatial resolution using bilinear interpolation. The normalized thermal and frequency priors are then concatenated

$$P = \mathrm{Concat}\left(\widehat{T}, \widehat{F}\right). \tag{9}$$

where T^ and F^ denote the normalized thermal and frequency responses, respectively.

Finally, a 1×1 convolution projects the two-channel prior representation to the desired number of feature channels:

$$\mathbf{F}_i^P = \mathrm{Conv}_{1\times1}\left(\left[\mathbf{F}_i^T, \mathbf{F}_i^F\right]\right), \tag{10}$$

where $\mathbf{F}_i^T$ and $\mathbf{F}_i^F$ denote the thermal and frequency-domain prior features, respectively, and $\mathbf{F}_i^P$ represents the resulting physics-aware prior feature map.

$$P_{\mathrm{out}} = \mathrm{Conv}_{1\times1}(P). \tag{11}$$

The resulting physics prior feature map ***Pout*** can then be fused with the original network features either by element-wise addition or by channel-wise concatenation.

3.3. Feature Restoration Enhancement

To enhance weak infrared Infrared small-target segmentation (IRSTS) at the deepest feature level, we introduce a **noise-guided feature refinement module**. However, the bottleneck layer contains only high-level semantic information extracted by the encoder and transformer. Infrared small targets only occupy a few pixels, so their responses may be weak after repeated downsampling, single head attention processing and feature extraction. We address this problem by introducing controlled Gaussian perturbations to bottleneck features during training and guiding their refinement using the previously extracted physics prior features. Let the bottleneck feature be represented as:

$$B \in \mathbb{R}^{C\times H\times W}. \tag{12}$$

$$P \in \mathbb{R}^{C_p \times H \times W}. \tag{13}$$

where C, H and W are the number of channels, height and width respectively and C_p is the number of prior channels. During training, Gaussian noise is injected into the bottleneck representation:

$$\widetilde{B} = B + \sigma_n \epsilon, \tag{14}$$

$$\epsilon \sim \mathcal{N}(0, I). \tag{15}$$

where ϵ denotes randomly sampled Gaussian noise and σ_n controls the strength of the noise as illustrated in Figure 2. In our implementation, σ_n is initialized to 0.2. The perturbed bottleneck feature is subsequently combined with the physics prior through channel-wise concatenation:

$$F_c = \mathrm{Concat}\left(\widetilde{B}, P\right). \tag{16}$$

The conditioned feature is then fed to a lightweight encoder with two 3×3 convolutional layers, followed by Group Normalization and ReLU. This process can be represented as:

$$\mathbf{F}_i^{(1)} = \mathrm{ReLU}\left(\mathrm{GN}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_i^{P}\right)\right)\right), \tag{17}$$

$$\mathbf{F}_i^{(2)} = \mathrm{ReLU}\left(\mathrm{GN}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_i^{(1)}\right)\right)\right), \tag{18}$$

where $\mathbf{F}_i^{P}$ denotes the conditioned physics-aware feature representation, $\mathbf{F}_i^{(1)}$ and $\mathbf{F}_i^{(2)}$ denote the intermediate feature maps, $\mathrm{Conv}_{3\times3}$ represents a 3×3 convolution, GN denotes Group Normalization, and ReLU represents the rectified linear unit activation function.

$$H = E_\theta\left(F_c\right). \tag{19}$$

where $E_\theta(\cdot)$ denotes the encoder and θ represents its learnable parameters. The encoder learns the joint relationship between the bottleneck representation and the physics prior, allowing it to distinguish target-relevant information from background responses.

The encoded representation is subsequently passed through a decoder to estimate a residual refinement feature:

$$R = D_\phi(H), \qquad R \in \mathbb{R}^{C \times H \times W}. \tag{20}$$

where $D_\phi(\cdot)$ denotes the decoder. The output R has the same number of channels and spatial dimensions as the original bottleneck feature. The refined bottleneck feature is finally obtained as

$$B_{\mathrm{out}} = B + \alpha R. \tag{21}$$

where α is a learnable residual scaling parameter initialized to 0.1. Importantly, the residual is added to the **original bottleneck feature** B rather than to the noisy representation $\widetilde{B}$. The complete training-time operation can be summarized as

$$B_{\mathrm{out}} = B + \alpha D_\phi\left(E_\theta\left(\mathrm{Concat}\left(B + \sigma_n \epsilon,\ P\right)\right)\right). \tag{22}$$

During inference, the Gaussian perturbation is removed and the clean bottleneck representation is directly combined with the physics prior:

$$B_{\mathrm{out}} = B + \alpha D_\phi\left(E_\theta(\mathrm{Concat}(B, P))\right). \tag{23}$$

Unlike conventional diffusion models **Ho et al. (2020)** that rely on multiple forward and reverse diffusion steps, the proposed approach performs a single-step noise-perturbation-guided feature refinement, substantially reducing computational complexity while retaining the denoising inspired principle of learning useful representations from corrupted features.

3.4. Edge Aware Efficient Transformer

To make full use of the fine-grained details and high-level semantic information extracted from all encoder levels, we exploit a single attention at the bottleneck. This mechanism can capture relationships between different spatial regions of the feature representation, allowing the model to learn broader contextual dependencies while preserving information relevant for small-target localization. We employ separable self-attention **Mehta and Rastegari (2023)** to the bottleneck of the network, which replaces the expensive token-to-token attention computation with element-wise operations and global context aggregation, reducing the complexity from quadratic to approximately linear with respect to the number of tokens.

Given an input feature sequence

$$X = [x_1, x_2, \dots, x_n]^{\mathsf{T}} \in \mathbb{R}^{n \times d}. \tag{24}$$

where d is the dimension and n is the number of patches, each token $\boldsymbol{x}$i is mapped to a scalar importance score through a learnable vector WI:

$$s_i = x_i^{\mathsf{T}} W_I. \tag{25}$$

A softmax operation normalizes these scores to determine the contribution of each patch. The resulting weights are used to aggregate the key features $\boldsymbol{K}$ into a global context vector:

$$a_i = \frac{\exp(s_i)}{\sum_{j=1}^{n} \exp(s_j)}, \tag{26}$$

$$g = \sum_{i=1}^{n} a_i K_i. \tag{27}$$

The global context is then combined with the value features $\boldsymbol{V}$ through element-wise multiplication:

$$y_i = V_i \odot g. \tag{28}$$

This allows important infrared target features to contribute more strongly to the global representation while reducing the influence of less informative background features. The transformed tokens are then reshaped into 2D feature map for dense prediction. Although separable self-attention has the ability to capture global information, the edges of infrared small targets are usually weak and ambiguous. Therefore, an **edge-aware module** **Cui et al. (2021)** is placed at each end of the transformer encoder stage to enhance edge-sensitive features.

$$G_l^x = I_l * K_x, \tag{29}$$

$$G_l^y = I_l * K_y. \tag{30}$$

where we represent the feature map at the ***l-th*** stage. The gradient magnitude is then calculated as

$$G_l(i,j) = \sqrt{\left(G_l^x(i,j)\right)^2 + \left(G_l^y(i,j)\right)^2}. \tag{31}$$

However, noise can also generate strong gradients. To use the local neighborhood, the gradient difference between a center pixel and its neighboring pixels is calculated as

$$\Delta_l(i,j;p,q) = \left|G_l(i,j) - G_l(p,q)\right|, \qquad (p,q) \in \mathcal{N}(i,j). \tag{32}$$

The standard deviation of these local gradient differences is then used to construct a local variation map $\boldsymbol{S}l$. Thus, $\boldsymbol{G}l$ measures the strength of spatial changes, while $\boldsymbol{S}$l measures their local variation.

Finally, adaptive thresholds are obtained from the mean values of the two maps. Candidate edge locations satisfying

$$G_l(i,j) > \tau_G, \tag{33}$$

$$S_l(i,j) > \tau_S, \tag{34}$$

$$\tau_G = \text{mean}(G_l), \tag{35}$$

$$\tau_S = \text{mean}(S_l). \tag{36}$$

are enhanced as

$$D_l(i,j) = \begin{cases} G_l(i,j), & G_l(i,j) > \tau_G \text{ and } S_l(i,j) > \tau_S, \\ 0, & \text{otherwise.} \end{cases} \tag{37}$$

while the remaining locations are suppressed to zero. This adaptive mechanism enhances meaningful target edges while reducing weak background responses and noise, complementing the global contextual information captured by separable self-attention. At the end original feature map $\boldsymbol{x}$i gets element wise multiplied with the enhanced map $\boldsymbol{Dl}$.

3.5. Boundary-Aware Segmentation Decoder

In each decoding stage, the low-resolution bottleneck features are progressively reconstructed and fused with the corresponding encoder features to restore the spatial details that may be attenuated during feature extraction. Given a bottleneck feature map:

$$F_b \in \mathbb{R}^{C_b \times H_b \times W_b}. \tag{38}$$

we first apply the inverse wavelet transform (IWT). The input feature Fb is decomposed into four wavelet sub-bands, which are the low and high-frequency components:

$$F_u = \text{IWT}\left(F_b^{LL}, F_b^{LH}, F_b^{HL}, F_b^{HH}\right). \tag{39}$$

Thus, IWT increases the spatial resolution by a factor of two while incorporating information retained in all four wavelet sub-bands.

However, directly concatenating the complete encoder feature with the decoder representation may also introduce irrelevant background responses. Therefore, before skip-feature fusion, spatial attention **Woo et al. (2018)** is applied to emphasize informative spatial locations. After spatial refinement, the attended encoder representation is concatenated with the IWT-upsampled decoder feature $\boldsymbol{Fu}$ along the channel dimension:

$$F_f = \text{Concat}\left(F_u, F_e^{\text{att}}\right). \tag{40}$$

A Sobel-based boundary attention mechanism is then further applied to Ff to strengthen local gradient information. For each channel of the fused feature map, horizontal and vertical gradients are independently calculated using the Sobel kernels Kx and Ky:

$$G_x = F_f * K_x, \tag{41}$$

$$G_y = F_f * K_y. \tag{42}$$

where denotes convolution. In the implementation, depth-wise convolution is employed so that the Sobel operation is independently applied to every feature channel rather than mixing information across channels.

The responses of the horizontal and vertical gradient are then combined to get the gradient magnitude as follows:

$$G = \sqrt{G_x^2 + G_y^2 + \varepsilon}. \tag{43}$$

A larger value of G here indicates a stronger local intensity variation and therefore a potentially more informative boundary location. The gradient magnitude is then converted into a soft boundary attention map using a sigmoid function,

$$A_b = \sigma(G). \tag{44}$$

The fused feature is subsequently reweighted using

$$F^F = F_f \odot A_b. \tag{45}$$

This enables the decoder to incorporate boundary-sensitive local information in addition to the spatially selected encoder features and the wavelet-reconstructed decoder representation.

Finally, the boundary-enhanced feature $\mathbf{F}^F$ is passed through two consecutive 3×3 convolutional layers. In the implemented stage, the first convolution transforms the feature representation from 128 to 64 channels. In this way, each decoding stage jointly exploits wavelet-based spatial reconstruction, spatially refined encoder information, and boundary-sensitive gradient cues, helping the network recover small infrared targets while reducing the influence of irrelevant background features.

3.6. Uncertainty Head

When dealing with Infrared small target segmentation, there are certain regions of an image which are inherently more ambiguous than others. For instance, pixels that are situated along target boundaries or near bright clutter sources (such as cloud edges or reflections) are difficult to classify with certainty, even under a well-trained model. This can lead to overconfident false positives in cluttered regions and unstable predictions along the target edges. To address this issue, we employ a heteroscedastic uncertainty estimation head that predicts both a segmentation map and a per-pixel uncertainty map. Given the final decoder feature map $F \in R^{B\times C\times H\times W}$, the two parallel convolutional branches computes the segmentation prediction and spatial uncertainty:

$$\mu(i,j) = f_{\text{mean}}(F)(i,j), \tag{46}$$

$$s(i,j) = f_{\text{var}}(F)(i,j) = \log \sigma^2(i,j). \tag{47}$$

Here, $\mu(i,j)$ is the predicted target logit and $s(i,j) = \log \sigma^2(i,j)$ is the predicted log-variance (uncertainty) at pixel location (i,j). Both branches have the same layer structure with a 3×3 convolution, Group Normalization, LeakyReLU activation, spatial dropout and a 1×1 convolution to project the features in a 1-channel map.

By incorporating this uncertainty-aware loss formulation, the network inherently starts to estimate its own spatial confidence without requiring any explicit supervision. The empirical observations demonstrate that the model systematically assigns elevated uncertainty values to ambiguous regions, such as target boundaries and high-intensity background clutter, while exhibiting high confidence (low uncertainty) across homogeneous background areas and distinct target centers.

3.7. Overall Training Objective

We combine **Focal Loss Lin et al. (2020)** and **Dice Loss Milletari et al. (2016)** to focus on difficult samples and directly optimize the overlap between the predicted and ground-truth target regions due to the severe imbalance between target and background in infrared small-target segmentation. They are defined as:

$$\mathcal{L}_{\text{focal}} = -(1-p_t)^{\gamma} \log(p_t), \tag{48}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon}. \tag{49}$$

where ***pt*** is the probability of the ground-truth class, γ is the focusing parameter, and ***pi*** and ***gi*** are the predicted probability and ground-truth label, respectively. The segmentation loss is formulated as:

$$\mathcal{L}_{\text{seg}} = \lambda_f \mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{dice}}. \tag{50}$$

where ***f***=10. By getting inspired from Lee et al. (2015), dice-based deep supervision is additionally applied to intermediate decoder outputs to improve target recovery at different feature levels. On the other hand infrared targets

may also contain low contrast, sensor noise, and ambiguous boundaries. To model such **aleatoric uncertainty**, the network predicts a mean logit ***i*** and log-variance ***si = logi2***, with

$$\tilde{z}_i^{(k)} = \mu_i + \exp\left(\frac{s_i}{2}\right)\epsilon_i^{(k)}, \qquad \epsilon_i^{(k)} \sim \mathcal{N}(0,1). \tag{51}$$

and then heteroscedastic BCE loss is obtained by averaging over K=10 Monte Carlo samples:

$$\mathcal{L}_{\text{het}} = \frac{1}{K}\sum_{k=1}^{K} \text{BCEWithLogits}\left(\tilde{z}^{(k)}, y\right), \qquad K = 10. \tag{52}$$

To discourage excessively large uncertainty, we use

$$\mathcal{L}_{\text{unc}} = \frac{1}{N}\sum_{i=1}^{N} \exp(s_i). \tag{53}$$

The final training objective is therefore:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{het}} + \lambda_u \mathcal{L}_{\text{reg}}. \tag{54}$$

where ***u*** =10-4. This objective jointly addresses severe class imbalance and observation-dependent uncertainty, encouraging accurate target localization while improving robustness to noisy and ambiguous infrared regions.

4. Experimental Setup

4.1. Datasets

We perform experiments on three commonly used public datasets: NUAA-SIRST, NUDT-SIRST and IRSTS-1K. These IRSTS benchmarks together demonstrate complementary characteristics such as object size, background noise. These set of single frame infrared images ensure PhyDiff U-Net to perform in a cross-domain robustness with varying environmental conditions.

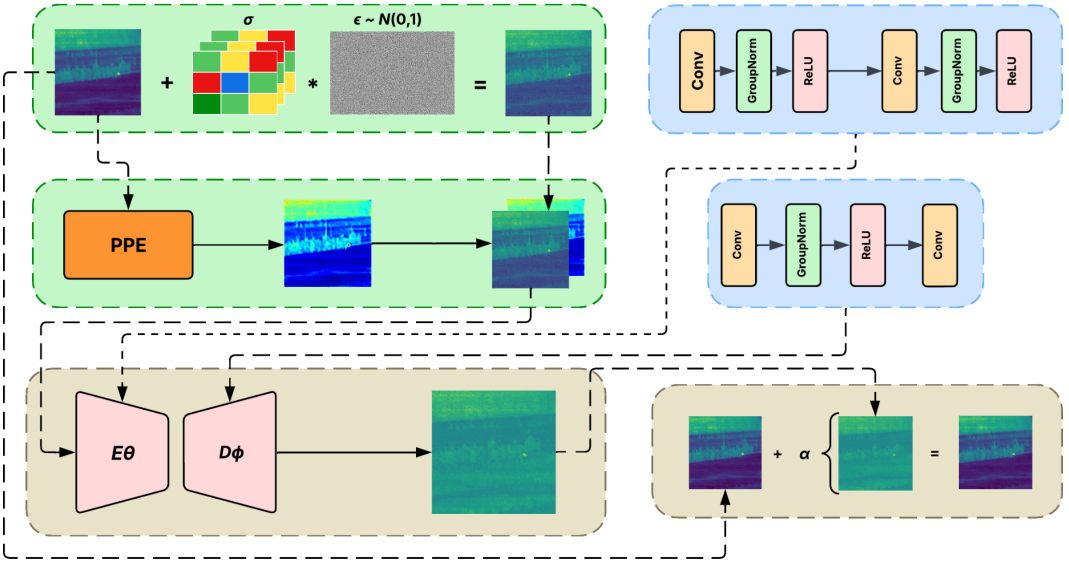


Figure 2: Noise-guided feature refinement module. Gaussian perturbation is applied to the bottleneck features during training, followed by physics-prior-conditioned encoding, residual refinement, and feature restoration.

NUAA-SIRST

In the dataset **NUAA-SIRST** Dai et al. (2021a), there are a total 427 real infrared images and among them 480 target instances are annotated that are collected from hundreds of distinct sequences. The images have a diverse set

Table 1
Summary of the infrared small-target datasets and experimental protocols used for training, validation, and evaluation.

Dataset	Type	Images	Target characteristics	Split Ratio
NUAA-SIRST	Real	427	2×2 to 14×34 pixels	90:5:5
NUDT-SIRST	Synthetic	1327	1×1 to 9×9 pixels	80:10:10
IRSTS-1K	Real	1000	1×1 to 56×33 pixels	80:10:10

of backgrounds including clouds, roads, mountains and rivers. Roughly 90% of the images have only one target and almost 35% of the targets are the brightest pixels in their respective images. It shows that if the saliency is based on brightness then it is not efficient for an accurate segmentation (Wang and Wang (2021), Kim et al. (2009)). The size range of the targets are 2×2 pixels to 14×34 pixels and the image dimensions range from 96×135 pixels to 400×592 pixels. The large imbalance in target and background makes it challenging to identify a negligible instance in an image in segmentation networks.

NUDT-SIRST

NUDT-SIRST Li et al. (2023a) has 1327 synthetic infrared images with the resolution of 256×256. The target size ranges from 1×1 pixels to 9×9 pixels. Unlike NUAA-SIRST, these images were generated by taking real-world infrared backgrounds and mathematically injecting synthetic small targets into them. This method helped overcome the scarcity of accurately labelled large-scale data for training Li et al. (2023a). A major characteristic of the NUDT-SIRST dataset is that only 35% of the targets are the brightest regions in the frame. The remaining 65% have pixel values blending into or lower than the surrounding clutter, such as heavy clouds or reflections of the ground. This prevents the model from detecting targets using just optimization of local brightness peaks. NUDT-SIRST is known for giving precise masks and the characteristics of a target in a much controlled way. In our study it has preserved extra small targets in PhyDiff U-net diffusion based restoration.

IRSTS-1K

In the third dataset that we have used, IRSTS-1K Zhang et al. (2022b), there are 1000 real infrared images of 512×512 resolution. It has scenes of mountains, rivers, oceans, wide open fields, cities and clouds. The dimensions of the target range from nearly 1×1 to 56×33 pixels covering both extra small and large size of targets Xi et al. (2025). As target categories there are instances of aircraft, vehicles, drones, boats, and other distant infrared objects. This dataset ensures generalization by focusing more on heterogeneous targets along with background than synthetic data. IRSTS-1K has both single-pixel and near to single-pixel targets which makes it even more relevant to our proposed physics conditioned model's super resolution stage.

In total, 2754 images have been evaluated in three datasets covering both real and synthetic instances in a wide range of target sizes. Studies of infrared small-target segmentation have established NUAA-SIRST , NUDT-SIRST, and IRSTS-1K as widely used public benchmarks.

4.2. Evaluation Metrics

The performance of an Infrared small-target segmentation (IRSTS) model can not be efficiently measured with a single metric. PhyDiff U-Net has used evaluation metrics like both pixel level segmentation and target level segmentation. Commonly used evaluation metrics in IRSTS for initial quantitative measurements are Intersection over Union (IoU), normalized Intersection over Union (nIoU), Probability of detection P_d and false-alarm rate $\boldsymbol{Fa}$. Detection performance evaluation metrics used in this study are receiver Operating Characteristic (ROC) curves.

1. Intersection over Union (IoU)

Intersection over Union ($\boldsymbol{IoU}$) is a metric to measure the overlap of two regions at the pixel level, the predicted target region and the corresponding ground truth region.

$$\text{IoU} = \frac{A_i}{A_u} = \frac{\sum_{i=1}^{N} TP[i]}{\sum_{i=1}^{N} (T[i] + P[i] - TP[i])}. \tag{55}$$

Here $\boldsymbol{Ai}$ is the intersection area and $\boldsymbol{Au}$ is the union regions. $\boldsymbol{N}$ is the total number of testing samples, $\boldsymbol{TP\ [i]}$ is the number of correctly predicted target pixels in the i-th sample, and T[i] and P[i] are the number of ground-truth target pixels and predicted target pixels, respectively. The spatial agreement between the predicted region and the ground-truth region is improved as $\boldsymbol{IoU}$ increases.

2. Normalized Intersection over Union (nIoU)

Often IoU gets dominated by the large background region because of the target containing a very small portion of the entire image. To normalize this possibility nIoU is applied in addition to a sample wise evaluation of target segmentation performance.

$$\text{nIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{TP[i]}{T[i]+P[i]-TP[i]}. \tag{56}$$

$\boldsymbol{N}$ is the total number of the test samples. The number of true-positive pixels for the i-th sample is denoted by TP[i]. T[i] is the number of ground truth pixels and P[i] is the number of predicted target pixels. With a higher nIoU value the accuracy of target localization and segmentation in complex background increases.

3. Probability of Detection (P_d)

Probability of Detection (P_d) is a target level metric that measures the performance of the individual infrared target detection. It is calculated as

$$P_d = \frac{N_{\text{pred}}}{N_{\text{all}}}. \tag{57}$$

Npred is the number of correctly detected targets, while ***Nall*** is the total number of ground-truth targets. In the widely adopted Infrared small target segmentation protocol, a predicted target is counted as correct if the Euclidean distance between the centroid of the predicted target and the corresponding ground-truth target is less than 3 pixels. If this condition is not satisfied then the target is considered missed.

$$\left\|C_{\text{pred}} - C_{\text{gt}}\right\|_2 = \sqrt{(x_{\text{pred}} - x_{\text{gt}})^2 + (y_{\text{pred}} - y_{\text{gt}})^2} < 3. \tag{58}$$

The above-mentioned formula is used as the target-matching criterion, where ***Cpred*** and ***gt*** represent the predicted and ground-truth target centroids respectively.

4. False Alarm Rate (***Fa***)

False Alarm Rate (***Fa***) is the amount of background that has been falsely classified as a target due to the structured clutter building target look alike. ***Fa*** is defined as the ratio of false predicted pixels to total number of pixels in the image:

$$F_a = \frac{N_{\text{false}}}{N_{\text{all}}}. \tag{59}$$

As the target-level matching protocol explains, a predicted response whose centroid deviation from the corresponding ground-truth target exceeds the 3-pixel threshold is considered an inaccurate detection. A lower value of F_a indicates better background suppression and fewer clutter-induced false alarms. In conventional IRSTS evaluation, F_a is typically reported in units of $\times 10^{-6}$ to represent very small false-alarm rates.

5. Receiver Operating Characteristic (***ROC***) Curve

ROC plots the model performance on a range of segmentation thresholds. It describes the relationship between the true-positive rate and false-positive rate as the decision threshold τ varies.

$$\text{TPR} = \frac{TP}{TP+FN}, \tag{60}$$

$$\text{FPR} = \frac{FP}{FP+TN}. \tag{61}$$

For different values of τ, the TPR and FPR are plotted to form the ROC curve. The trade-off between missed detections and false detections can also be described using the detection probability P_d and false-alarm rate F_a. Closer the curve is to the upper left region the better the performance in target detection.

4.3. Implementation Details

All input infrared images were converted to single-channel grayscale and resized to 256×256 pixels, except for NUDT-SIRST images, which are already provided at a resolution of 256×256 pixels. The proposed network does not employ super-resolution and operates at the native input resolution, using DWT and IWT for downsampling and upsampling, respectively. The noise-perturbation-guided feature refinement follows a single-step mechanism, where Gaussian noise with a fixed standard deviation of $\boldsymbol{\sigma} = 0.20$ is added only during training, while inference requires a single forward pass without noise injection. The Transformer bottleneck operates on a 32×32 feature map with an embedding dimension of 1024 tokens, one attention head, one Transformer block, a feed-forward expansion ratio of 4, and a dropout rate of 0.1. The PPE modules use a Gaussian PSF with $\boldsymbol{\sigma} = 1.0$ and a 7×7 kernel, an annular filter with $r_{\text{in}} = 3$ and $r_{\text{out}} = 9$, and Haar wavelet decomposition to extract thermal-contrast and high-frequency priors. Our model was trained for 100 epochs using Adam optimizer with an initial learning rate of 3×10^{-4}, weight decay of 10^{-4} and batch size of 1. A ReduceLROnPlateau scheduler was employed to reduce the learning rate by a factor of 0.8 after five epochs without improvement with a minimum learning rate of 10^{-6}. Random flipping and rotations between $-10°$ and $+10°$ were applied during training. All experiments were conducted on Google Colab using a single NVIDIA L4 GPU with 24 GB memory. A fixed seed of 42 was used for dataset splitting, while no global seed was set for model initialization or other stochastic training operations. NUAA-SIRST contains substantially fewer samples than NUDT-SIRST and IRSTS-1K, we allocate a larger proportion of its images to training to improve training stability. Accordingly, NUAA-SIRST is divided using a 90:5:5 train-validation-test ratio, whereas NUDT-SIRST and IRSTS-1K use an 80:10:10 split. For each dataset, the same partition is retained across all compared methods and ablation experiments to ensure fair evaluation. Configuration was trained only once for each dataset and the checkpoint with highest validation IoU was used for final evaluation.

5. Results and Analysis and Discussion

5.1. Comparison with State-of-the-Art Methods

Quantitative comparisons with state-of-the-art methods are presented in Table 2. Overall, the proposed PhyDiff U-Net demonstrates highly competitive performance across the four evaluation metrics, namely IoU, nIoU, P_d, and Fa, achieving either the best or second-best results in most cases. Even when it does not obtain the highest score, the performance gap with the leading neural networks remains relatively small. Notably, PhyDiff U-Net achieves the best performance across all four metrics on the IRSTS-1K dataset, with an IoU of 0.6575, nIoU of 0.6598 and P_d of 0.9283. Compared with UIU-Net, the second-best method in terms of segmentation accuracy, PhyDiff U-Net improves IoU and nIoU by 0.0006 and 0.0169, respectively. More importantly, proposed method obtains the best P_d and the minimum Fa on NUAA-SIRST and NUDT-SIRST simultaneously, which demonstrate the capability of the proposed method in localizing infrared small targets accurately and suppressing false detections effectively.

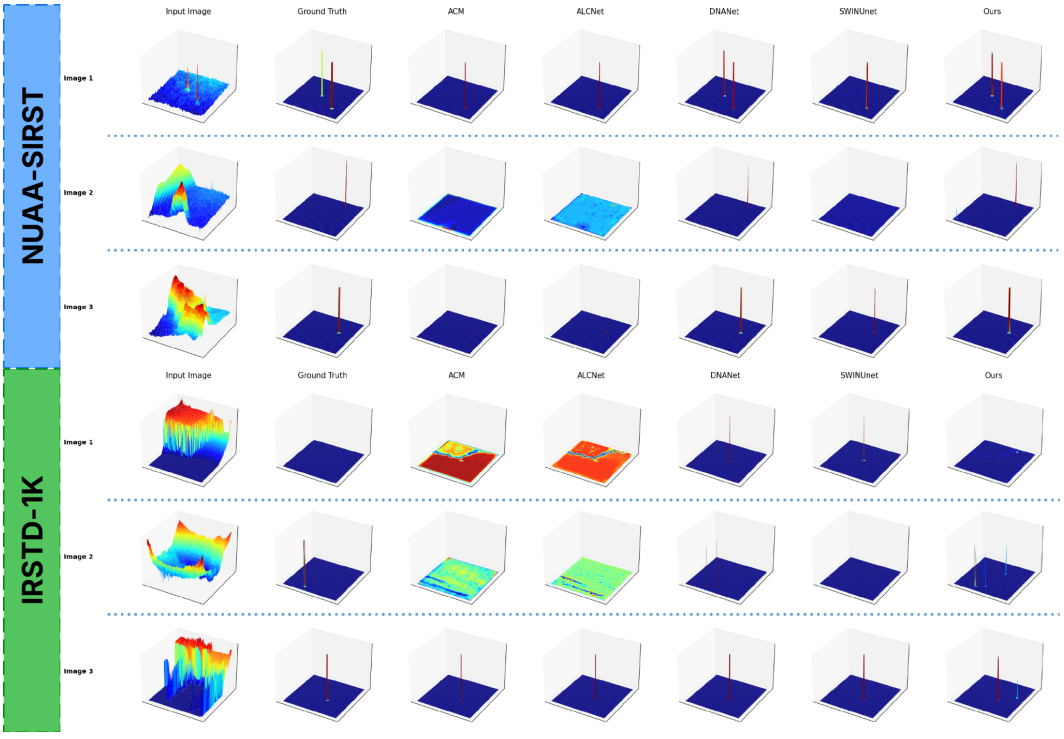


Figure 3: Three-dimensional visualization of infrared inputs, ground-truth masks, and predicted saliency responses produced by different methods on the NUAA-SIRST and IRSTS-1K datasets. The x-axes and y-axes correspond to the spatial coordinates of each image, while the z-axis represents the response magnitude at each pixel. For the input images, the height indicates infrared intensity, for the ground-truth masks, it represents the binary target label, and for the model outputs, it represents the predicted target response, where larger values indicate stronger target confidence.

On NUAA-SIRST, PhyDiff U-Net achieves the second-highest IoU of 0.7687 while obtaining the highest P_d of 0.9916 and the lowest Fa of 3.34. Similarly, on NUDT-SIRST, the proposed model achieves competitive IoU and nIoU values of 0.9120 and 0.9239, respectively, while producing the lowest false-alarm rate of 12.93. Although UIU-Net achieves a slightly higher P_d on NUDT-SIRST (0.99 compared with 0.98), the difference is only 0.01, whereas PhyDiff U-Net substantially reduces the false-alarm rate. These results indicate that the proposed method provides a strong balance between pixel-level segmentation accuracy, target segmentation capability, and false-alarm suppression. All the competing models were trained for the same number of epochs in an effort to provide a fair comparison. In this fixed training setting, the proposed PhyDiff U-Net converges faster than the other methods, which shows that it is more efficiently optimized during training.

As illustrated Figure 7, the proposed PhyDiff U-Net shows a favorable TPR–FPR trade-off compared with the competing learning-based methods across the evaluated datasets. In particular, on the challenging IRSTS-1K dataset Figure 7, PhyDiff U-Net maintains a relatively high true-positive rate while producing fewer false-positive responses over a wide range of thresholds. This advantage remains evident even at relatively low thresholds, where models generally become more sensitive to potential targets and consequently tend to generate more false positives. For example, at a threshold of 0.3, PhyDiff U-Net achieves strong target sensitivity while maintaining a comparatively low false-positive rate, indicating that the model can effectively distinguish small infrared targets from complex background clutter.

On NUAA-SIRST Figure 7, ALCNet exhibits an ROC behavior comparable to that of PhyDiff U-Net over part of the operating range. Nevertheless, the proposed model remains highly competitive in terms of the overall TPR–FPR balance. The reduced separation between the methods on NUAA-SIRST may be influenced by differences in dataset characteristics, including its relatively limited sample size and background diversity. Overall, the ROC analysis demonstrates that PhyDiff U-Net provides robust target discrimination across different decision thresholds while effectively controlling false-positive rate.

Table 2
Quantitative comparison of PhyDiff U-Net with representative infrared small-target segmentation methods on NUAA-SIRST, IRSTS-1K, and NUDT-SIRST. All competing models were reproduced under the unified experimental protocol. The best and second-best results are shown in bold and underlined, respectively.

Model	NUAA-SIRST				IRSTS-1k				NUDT-SIRST			
	IoU↑	nIoU↑	P_d↑	Fa↓	IoU↑	nIoU↑	P_d↑	Fa↓	IoU↑	nIoU↑	P_d↑	Fa↓
U-Net Ronneberger et al. (2015)	0.51	0.56	0.87	52.34	0.60	0.47	0.70	44.23	0.09	0.61	0.81	61.76
ACM-U-Net Dai et al. (2021a)	0.64	0.62	0.95	17.37	0.60	0.59	0.90	63.26	0.82	0.83	0.98	39.73
DNANet Li et al. (2023a)	0.74	0.74	0.87	9.75	0.61	0.61	0.77	**14.22**	**0.94**	**0.93**	0.96	24.85
Swin-U-Net Cao et al. (2022)	0.58	0.58	0.73	33.87	0.46	0.46	0.61	25.67	0.64	0.64	0.73	23.78
UIU-Net Wu et al. (2023b)	**0.77**	**0.75**	0.92	8.23	**0.65**	0.64	0.91	22	0.90	0.89	**0.99**	19.53
ALCNet Dai et al. (2021b)	0.64	0.67	0.93	42.56	0.62	0.60	0.87	58.80	0.69	0.69	0.94	38.75
PhyDiff U-Net	0.76	0.73	**0.99**	**3.34**	**0.65**	**0.65**	**0.92**	16.63	0.91	0.92	0.98	**12.93**

5.2. Main Component Ablation

PhyDiff U-Net, U-Net is used as baseline to evaluate the effectiveness of the proposed components. The baseline U-Net performs poorly on the NUAA-SIRST dataset, achieving an IoU of 0.5181, nIoU of 0.5600, P_d of 0.8722, and a relatively high false-alarm rate of 0.4410, as shown in Table 2. After incorporating the proposed modules into the network, the overall segmentation and detection performance improves significantly, demonstrating the effectiveness of the proposed architecture.

PhyDiff U-Net w/o EET and FRE: To further investigate the contribution of individual components, Table 3 presents an ablation study in which one or more modules are removed while the remaining components are retained. For example, when EET and FRE are simultaneously removed, the PPE, DWT, and SDA modules remain active in the architecture. Removing both EET and FRE reduces the model size from 3.16M to 1.47M parameters, corresponding to a reduction of approximately 1.69M parameters. However, this reduction in complexity is accompanied by a decrease in performance. IoU on NUAA-SIRST decreases from 0.7687 to 0.7530, and on NUDT-SIRST decreases from 0.9120 to 0.8800. EET and FRE also help to improve the overall representation and segmentation capability of the network, as evidenced by similar reductions in nIoU and P_d.

PhyDiff U-Net with or w/o EET and FRE: A more pronounced degradation is observed when EET or FRE is removed individually. Without EET, the IoU and nIoU on NUAA-SIRST decrease to 0.7030 and 0.6890, respectively, whereas removing FRE further reduces them to 0.6950 and 0.6680. Compared with the complete model, removing EET results in decreases of 0.0657 in IoU and 0.0489 in nIoU, while removing FRE results in decreases of 0.0737 in IoU and 0.0699 in nIoU on NUAA-SIRST. A similar, although smaller, performance reduction is observed on NUDT-SIRST. Performance of the FRE module in the proposed architecture is also illustrated by the feature maps in Figure 5. Although EET enhances long-range semantic dependencies between spatial regions through the separable self-attention mechanism, repeated downsampling and global feature modeling may lead to the loss of fine-grained local information, particularly the weak target responses and boundary details that are crucial for infrared small-target segmentation. Consequently, while EET provides strong contextual information and improves the semantic representation of the target relative to the surrounding background, this information alone may not be sufficient for the decoder to accurately reconstruct very small target regions. To address this limitation, FRE is introduced at the bottleneck to further refine the target features. Specifically, FRE inserts the bottleneck features using Gaussian noise and subsequently restores the informative target features under the guidance of the physics-prior representation. This refinement process helps suppress irrelevant background responses while recovering target-specific details that may have been weakened during encoding and attention-based feature modeling. Therefore, EET primarily strengthens global contextual relationships, whereas FRE complements it by refining and restoring fine-grained target information. Their joint integration significantly contributes to the stronger segmentation performance of the complete PhyDiff U-Net.

PhyDiff U-Net w/o PPE: In contrast, removing PPE results in a comparatively smaller degradation. The model without PPE achieves an IoU of 0.7570 on NUAA-SIRST compared with 0.7687 for the complete model, while the parameter count decreases only slightly from 3.16M to 3.15M. On NUDT-SIRST, removing PPE reduces the IoU from 0.9120 to 0.9000. Interestingly, the nIoU on NUAA-SIRST increases slightly from 0.7379 to 0.7420 after removing

Table 3
Ablation study evaluating the contributions of the principal components of PhyDiff U-Net. Values are reported as IoU / nIoU / P_d.

Model	Params (M)	Datasets	
		NUAA-SIRST	NUDT-SIRST
PhyDiff U-Net	3.16	0.76 / 0.73 / 0.99	0.91 / 0.92 / 0.98
PhyDiff U-Net w/o EET and FRE	1.47	0.75 / 0.73 / 0.99	0.88 / 0.90 / 0.96
PhyDiff U-Net w/o EET	1.71	0.70 / 0.68 / 1.00	0.89 / 0.91 / 0.97
PhyDiff U-Net w/o FRE	2.91	0.69 / 0.66 / 0.95	0.90 / 0.91 / 0.98
PhyDiff U-Net w/o PPE	3.15	0.75 / 0.74 / 1.00	0.90 / 0.91 / 0.98
PhyDiff U-Net w/o Uncertainty Head	3.15	0.73 / 0.68 / 0.97	0.92 / 0.94 / 0.97
PhyDiff U-Net w/o DWT	2.66	0.77 / 0.72 / 1.00	0.92 / 0.93 / 0.98
PhyDiff U-Net w/o IWT	3.24	0.74 / 0.70 / 0.98	0.92 / 0.93 / 0.98

PPE, indicating that the contribution of PPE may vary across individual evaluation metrics. Nevertheless, as results depicted in last row of Table 3, PPE is primarily designed to emphasize regions with strong local thermal contrast and high-frequency responses, which are important cues for identifying infrared small targets. Since the intensity of a target center often differs noticeably from its surrounding neighborhood, the local contrast prior helps highlight potential target regions. Meanwhile, the high-frequency wavelet components especially the LH, HL, and HH sub-bands retain structural details such as target boundaries, shapes, and fine textures, while reducing the influence of less informative background regions. As visualized in Figure 6, PPE assigns greater importance to target-specific regions than to the surrounding background. In particular, for the second example in Figure 6, the PPE response clearly distinguishes the small target from complex background structures such as grass. This suggests that the physics-prior representation can improve target–background discrimination and provide more informative features for detecting small targets with varying shapes and sizes. Nevertheless, the complete PhyDiff U-Net provides the strongest overall balance across datasets and evaluation criteria, suggesting that the proposed modules contribute complementary information to the final architecture.

PhyDiff U-Net w/o Uncertainty Head: On NUAA-SIRST, there is a noticeable amount of reduction in performance once the uncertainty estimation head is removed. The IoU of the model was noted 0.7687 and nIoU 0.7379, which reduces to IoU 0.7300 and nIoU 0.6800 upon the removal of the uncertainty head. Uncertainty estimation does not only take part in confidence estimation but also helps to increase the reliability of target localization. The value of P_d also decreases to 0.9700 in the absence of uncertainty head. For the NUDT-SIRST dataset, these values are seen to be at a decreased state as IoU of 0.9200, nIoU of 0.9400, and P_d of 0.9700. With ambiguous target boundaries and if the background is structured in bright surroundings, the predictive results come out as weak. This situation is benefitted with observation dependent uncertainty.

PhyDiff U-Net w/o DWT: If we remove the DWT downsampling on NUAA-SIRST, the model reaches to a parameter count to 2.66M with an IoU of 0.7700, nIoU of 0.7200, and P_d of 1.0000. On NUDT-SIRST these values are IoU 0.9200, nIoU 0.9300, and P_d 0.9800. The value of IoU is a little bit higher on NUAA-SIRST than the complete model, but the value of nIoU decreases from 0.7379 to 0.7200. This shows that the DWT implementation is useful in providing high and low frequency information rather than discarding fine details during downsampling.

PhyDiff U-Net w/o IWT: Removing IWT upsampling, the number of parameters increases to 3.24M, and the standard metrics for performance drop to an IoU of 0.7400 and nIoU of 0.7000 on NUAA-SIRST, with a P_d of 0.9800. The model generates NUDT-SIRST results of IoU 0.9200, nIoU 0.9300 and P_d 0.9800. This reduction implies the weak recovery of target details compared with the full model, due to the lack of inverse wavelet directly reconstructing the spatial resolution. The frequency-preserving encoder-decoder path is implemented by combining the IWT and the DWT.

Additional Comparative Analysis: In addition, we also compare representative methods on NUAA-SIRST with 80:10:10 train-validation-test split, to make the comparison more consistent. As shown in Table 7, PhyDiff Unet

achieves competitive performance for all three metrics: IoU is 0.6867, nIoU is 0.6930 and P_d is 0.9327, which are the second best results. The proposed UIU-Net shows the best overall segmentation and detection performance. PhyDiff Unet can segment target well and gets F_a of 11.54. Results also demonstrate the efficiency of the proposed framework using the same data split.

Table 4
Analysis of key design choices for physics-prior composition, Edge Aware Efficient Transformer and the Feature Restoration Enhancement on the NUAA-SIRST dataset.

Component	Configuration	Variant	IoU ↑	P_d ↑
Physics Prior Extraction (PPE)	Prior Type	Thermal only	76.16	96.82
		Frequency only	71.85	95.23
		Thermal + Frequency	76.87	99.16
Edge-Aware Efficient Transformer (EET)	Transformer Depth	1	76.87	99.16
		2	72.36	99.87
		3	75.15	96.82
	Attention Heads	1	76.87	99.16
		2	74.64	99.85
		4	71.78	96.82
Feature Restoration Enhancement (FRE)	With Prior Tensor		76.87	99.16
	Without Prior Tensor		70.24	98.23

5.3. Key Design Analysis

Beyond evaluating the contribution of each individual module, we further investigate how different design choices within these modules affect performance. Specifically, we examine the PPE module using thermal information alone, frequency information alone, and their combination. We also analyze different configurations of the EET and FRE modules to better understand how their internal design choices influence overall segmentation performance.

5.3.1. Analysis of PPE

The PPE module is designed to reduce the network's reliance on target size, which can vary considerably across infrared images. Instead, it exploits two relatively consistent characteristics of infrared small targets, local thermal contrast and frequency information. Small targets are generally compact and exhibit higher intensity than their surrounding background, while their localized structure also produces distinctive frequency responses.

As shown in Table 4, using only the thermal prior results in a relatively small decrease in IoU, from 76.87% to 76.16%, corresponding to a drop of only 0.71%. However, P_d decreases from 99.16% to 96.82%, indicating that some targets are missed when frequency information is removed. In contrast, using only the frequency prior causes a more substantial drop, with IoU decreasing to 71.85% and P_d to 95.23%, representing drops of 5.02 and 3.93%, respectively. This suggests that frequency information alone is insufficient to reliably distinguish small targets from complex background structures. Thermal contrast appears to provide the stronger cue for target localization, while frequency information provides complementary structural and detail information. Combining both priors therefore produces the best overall performance, demonstrating their complementary roles in preserving target responses and suppressing background interference.

5.3.2. Analysis of EET

The EET module is introduced to capture the relationship between potential target regions and their surrounding background. For infrared small-target segmentation, a target cannot always be identified reliably from its local appearance alone, understanding its surrounding context is also important. The transformer therefore provides global semantic information by modeling relationships between different image regions, while the edge-aware component introduces additional local boundary information to help preserve the fine structure of small targets.

Table 4 shows that increasing the transformer complexity does not necessarily improve segmentation performance. With only one transformer layer, the model yields an IoU of 76.87% and P_d of 99.16%. Increasing the depth to two layers slightly increases P_d to 99.87%, but decreases IoU significantly to 72.36%. A similar trend is observed for the

number of attention heads: using two heads increases P_d to 99.85%, but decreases IoU from 76.87% to 74.64%, while four heads further reduce IoU to 71.78%. These results suggest that excessive transformer processing may over-mix spatial information, which is undesirable for extremely small targets whose representations occupy only a few pixels. Although additional depth or attention heads can occasionally improve target segmentation, they may also weaken precise target localization and boundary preservation. Therefore, the single-layer, single-head configuration provides a more balanced representation of global context and local target structure while maintaining lower model complexity.

5.3.3. Analysis of FRE

For the FRE module, we analyze whether the physics-prior tensor is actually useful during feature refinement. As shown in Table 4, using the prior tensor gives an IoU of 76.87% and a P_d of 99.16%. When the prior tensor is removed, the IoU drops to 70.24%, a decrease of 6.63%, while P_d decreases slightly to 98.23%.

This result shows that the physics prior mainly helps the model produce a more accurate target region rather than only improving whether a target is detected or not. Without the prior tensor, the FRE module can still detect most targets, as indicated by the relatively high P_d, but its segmentation quality becomes much worse. This suggests that the prior tensor provides useful information about the likely target location and structure, helping the FRE module refine weak target features and separate them more clearly from the surrounding background.

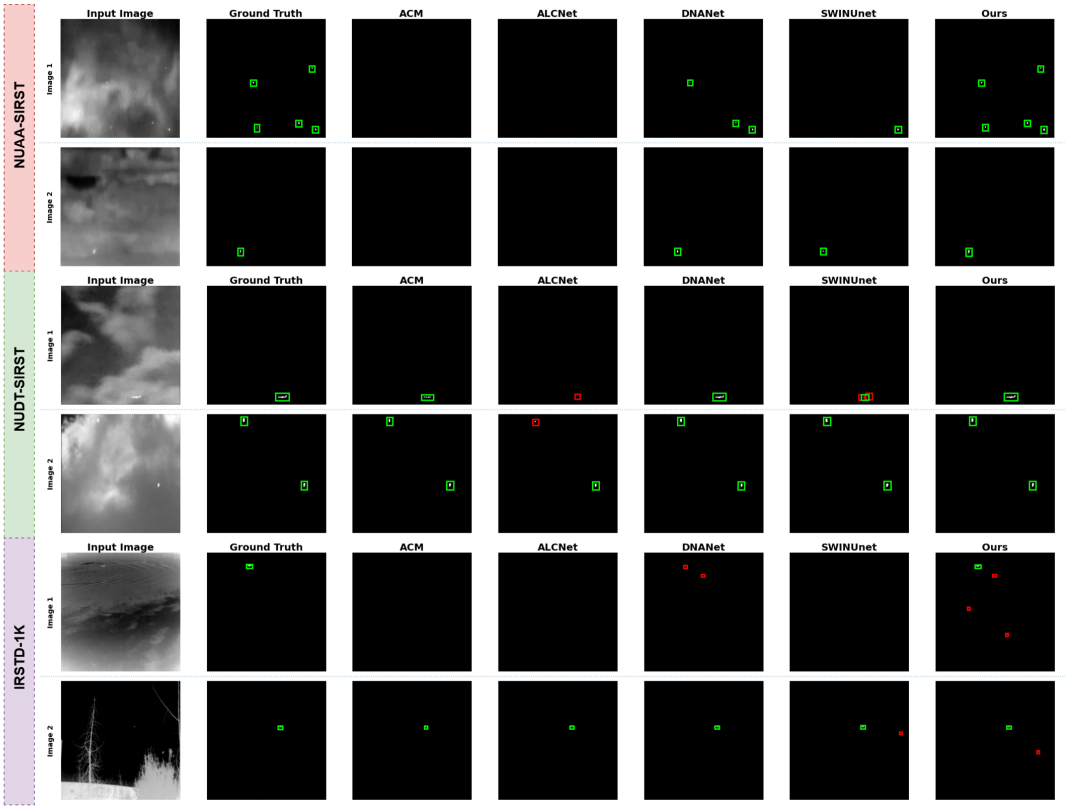


Figure 4: Qualitative comparison of PhyDiff U-Net with representative state-of-the-art methods on challenging infrared satellite scenes. Examples are very small targets, low contrast targets, structured background clutter, strong edges and multiple target conditions.

5.4. Robustness and Generalization

To assess the robustness and generalization capacity of the proposed PhyDiff U-Net, we perform additional experiments in two difficult scenarios, i.e., different target-size ranges and cross-dataset evaluation. For the target-size analysis, samples from the NUDT-SIRST dataset are divided into two groups containing targets of 1–3 pixels and 4–9 pixels, respectively. As shown in Table 5, the proposed model maintains strong segmentation and detection performance across both target-size ranges.

Table 5
Robustness and cross-dataset generalization of PhyDiff U-Net across target sizes, observation and domain-shift conditions.

Evaluation	Condition	Baseline IoU	Ours IoU	Ours P_d
Target size	1–3 px	68.10	98.83	99.68
Target size	4–9 px	4.82	84.69	95.55
Cross-dataset	NUDT→NUAA	5.58	49.85	92.85
Cross-dataset	NUAA→NUDT	6.66	42.41	84.74

To evaluate the robustness and generalization capability of the proposed PhyDiff U-Net, we conduct additional experiments under two challenging conditions: different target-size ranges and cross-dataset evaluation. For the target-size analysis, samples from the NUDT-SIRST test set were divided into two groups based on target size: 1-3 pixels and 4-9 pixels, comprising 64 and 69 images, respectively. The groups were constructed without any data leakage. As shown in Table 4, the proposed model maintains strong segmentation and detection performance across both target-size ranges. PhyDiff U-Net achieves an IoU of 98.83% and a P_d of 99.68% for very small targets of 1–3 pixels, which is greatly better than the IoU of 68.10% of the baseline. The result shows the effectiveness of the proposed architecture in preserving the spatial structure of very small infrared targets and separating them accurately from the background. For 4-9 pixel targets, PhyDiff U-Net achieves 84.69% IoU and 95.55% P_d, while the baseline achieves only 4.82% IoU. The segmentation IoU is a little lower than that achieved for the 13 pixel subset, but the model still maintains high target-level detection accuracy and pixel-level segmentation accuracy. These results demonstrate the proposed method is able to deal with infrared targets of various spatial structures.

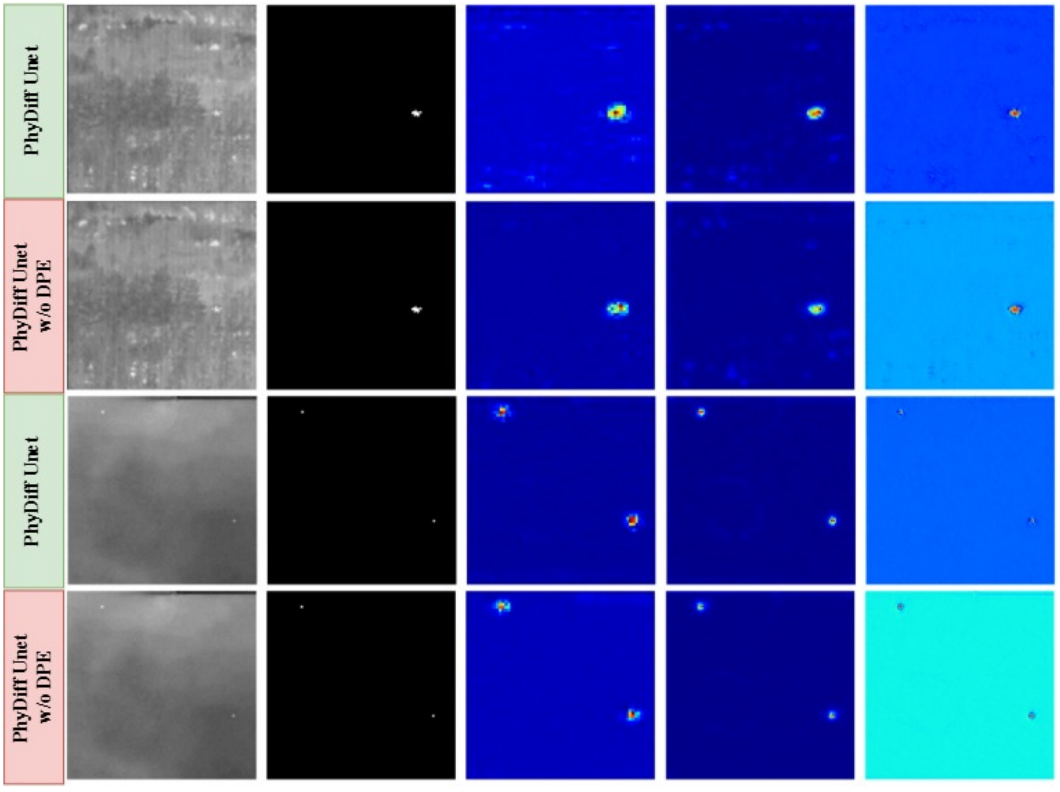


Figure 5: Feature maps of Phydiff U-Net and Phydiff U-Net without FRE The representation map validates the effectiveness of the module by assigning a significantly higher strength to the specific target points in case of small targets rather than the regions.

We further evaluate cross-dataset generalization to investigate the robustness of the model under domain shift. When trained on NUDT-SIRST and evaluated on NUAA, PhyDiff U-Net achieves an IoU of 49.85% and a P_d of

92.85%, substantially higher than the baseline IoU of 5.58%. Similarly, when trained on NUAA and evaluated on NUDT-SIRST, the proposed model achieves an IoU of 42.41% and a P_d of 84.74%, compared with a baseline IoU of only 6.66%. Although the segmentation accuracy decreases compared with the within-dataset evaluation, which is expected because of differences in target appearance and background distributions between datasets, the relatively high P_d values indicate that the proposed model retains a strong ability to localize small infrared targets under domain shift. Overall, these experiments demonstrate that PhyDiff U-Net provides improved robustness across different target sizes and better cross-dataset generalization than the baseline model.

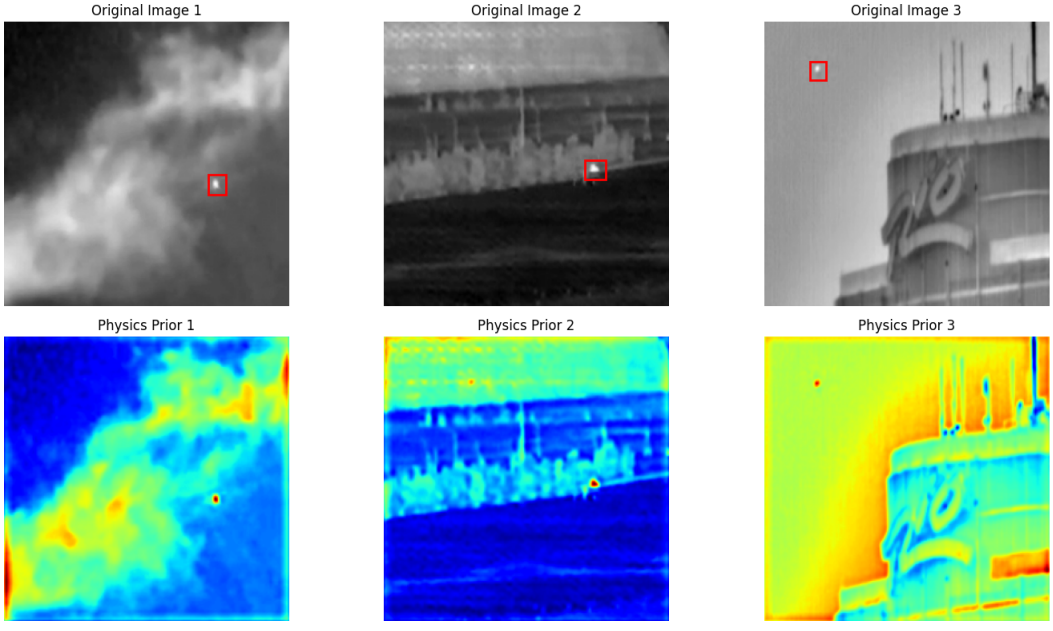


Figure 6: Visualization of physics-prior representations generated by the Physics Prior Extraction Module (PPE) during the encoding stage. The extracted priors focus on high-response target regions while preserving informative structural characteristics of the surrounding background. Red regions indicate high-response areas.

5.5. Complexity and Sampling Efficiency

The computational comparison in Table 6 further demonstrates the efficiency of the proposed architecture. PhyDiff U-Net contains 3.16M parameters and requires 5.732G FLOPs while achieving an IoU of 76.87%. In comparison, UIU-Net achieves an IoU of 77.53%, but requires 50.54M parameters and 54.426G FLOPs. Moreover, PhyDiff U-Net operates at 151.39 FPS with a latency of 6.605 ms, compared with 39.45 FPS and 42.6 ms for UIU-Net. Thus, PhyDiff U-Net achieves relatively better inference speed while using substantially fewer parameters and computations. Although lightweight methods such as ACM-Net contain fewer parameters and achieve higher inference speeds, the showcased IoU is 64.01%. The results demonstrate a favorable trade-off between segmentation accuracy, target-level segmentation performance, false-alarm suppression, computational complexity, and inference efficiency, making it suitable for resource-constrained and real-time infrared small-target segmentation applications.

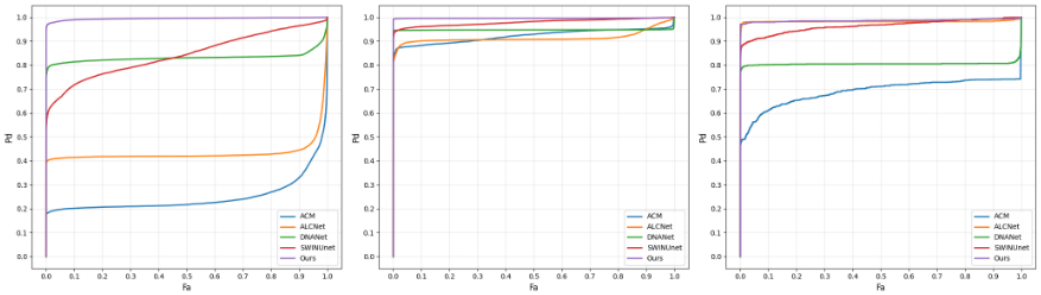


Figure 7: ROC curves of various deep-learning methods on IRSTS-1K, NUDT-SIRST and NUAA-SIRST datasets. The proposed PhyDiff model can achieve high detection probability (P_d) with very low false-alarm rate (Fa) on IRSTS-1K and NUDT-SIRST. Meanwhile, ALCNet achieves similar performance with PhyDiff on NUAA-SIRST.

To evaluate the practical deployability of the proposed PhyDiff U-Net, we compare the segmentation accuracy (IoU) with the computational cost (FLOPs) of all competing methods in Fig. 8. Heavyweight models such as UIU-Net (50.54 GFLOPs, 54.42M parameters) and DNANet (14.261 GFLOPS, 4.698M parameters) achieve competitive IoU scores, but require much higher computational resources. In contrast, lightweight models such as ACM-U-Net (23.841 GFLOPs) and ALCNet (0.378 GFLOPs) significantly reduce computational complexity, but their segmentation accuracy is lower. PhyDiff U-Net provides a better balance between accuracy and efficiency. In particular, it achieves the highest IoU on the IRSTS-1K dataset while maintaining a moderate computational cost.

Table 6
Computational complexity and inference efficiency of PhyDiff Unet on NUAA SIRST compared with representative competing methods.

Method	Params (M) ↓	FLOPs (G) ↓	FPS ↑	Latency ↓	IoU ↑
ACM-Unet	0.40 M	23.841G	328.84	3.041 ms	64.01
DNA-Net	4.697 M	14.261G	86.68	11.537 ms	74.60
Swin-UNet	27.025 M	7.44G	92.27	10.838 ms	58.70
ALCNet	0.427 M	0.378G	252.72	3.957 ms	64.35
UIUnet	50.54 M	54.426G	39.45	42.6 ms	77.53
PhyDiff Unet	3.16 M	5.732G	151.39	6.605 ms	76.87

5.6. Failure-Case Analysis

Despite the overall effectiveness of the proposed PhyDiff U-Net, several challenging cases reveal limitations of the model, particularly in scenes containing strong background clutter and target-like structures. As shown in Figure 4, the proposed model may occasionally either generate false alarms or fail to identify extremely weak targets. In Figure 4, PhyDiff U-Net successfully detects the ground-truth target, however, it also incorrectly identifies three additional background regions as targets. This indicates that complex background structures with local responses similar to those of infrared small targets can still mislead the model, resulting in false alarms. Although the proposed architecture incorporates contextual and physics-guided information to suppress background interference, highly cluttered scenes remain challenging when several background regions exhibit target-like intensity or structural characteristics.

A different failure mode can be observed in Figure 4, where the proposed model fails to detect the ground-truth target. In this example, the target is surrounded by strong background structures, including leaves with local intensity and texture characteristics that may reduce the contrast between the target and its surroundings. Consequently, the target response may be suppressed during feature extraction, leading to a missed segmentation. The corresponding 3D response visualization in Figure 3 further shows that the target location receives a comparatively weak response, suggesting that the target representation is insufficiently emphasized relative to the surrounding clutter. This observation suggests that, although the proposed model effectively captures small-target information in most cases, further improvement in discriminating extremely weak targets from structurally similar clutter could enhance its robustness under particularly challenging conditions.

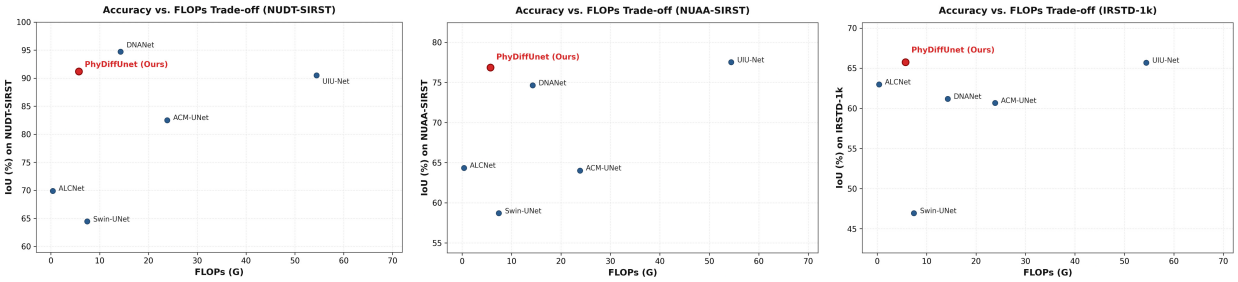


Figure 8: The accuracy efficiency trade-off between IoU and FLOPs of different methods on (i) NUDT-SIRST, (ii) NUAA-SIRST, and (iii) IRSTS-1K. The PhyDiff U-Net achieves a good trade-off between segmentation accuracy and computational cost for all three datasets.

5.7. Discussion

The acquired results from the experiments demonstrates PhyDiff U-Net providing a balance between correctness of pixel-level segmentation, prediction accuracy of target-level, reducing false alarm rate and calculative efficiency. The achievements collected from NUAA-SIRST, NUDT-SIRST, and IRSTS-1K contribute to the approach of physics-prior extraction, efficient transformer, diffusion-based approach, redevelopment based on wavelet, boundary aware decoding and assumption of uncertainty. In this section, we discussed further insights about the contribution of the PhyDiff U-Net model.

Table 7
Additional quantitative evaluation on NUAA-SIRST using an 80:10:10 train-validation-test split for a fair and consistent comparison.

NUAA-SIRST				
Model	**IoU** ↑	**nIoU** ↑	P_d ↑	**Fa** ↓
ACM-Unet	0.1153	0.1162	0.3173	22.43
DNANet	0.6148	0.5730	0.8710	4.74
Swin-UNet	0.4263	0.3668	0.7358	21.46
UIU-Net	**0.7457**	**0.7315**	**0.9853**	**1.83**
ALCNet	0.1250	0.1127	0.3077	26.45
PhyDiff Unet	0.6867	0.6930	0.9327	11.54

5.8. Why Physics Priors Help

One of the prior inspirations of our proposed model is that it combines physical and structural components that are inhabited in Infrared Small Targets. These characteristics can be utilized initially in the process of semantic feature extraction. In the proposed Physics Prior Extraction (PPE) module, thermal contrast and frequency representation have been incorporated. The thermal prior calculates the difference between a PSF weighted local response and its background. This speculation is particularly beneficial in extremely small target scenarios that can be distinguished from its background locally.

5.9. Role of Feature Restoration Enhancement

In a repetitive downsampling technique and transformer processing weak target representations are strengthened by diffusion inspired bottleneck enhancement. While passing through an encoder, an infrared image gets squeezed into a bottleneck which is a space of low resolution and high channel. If the target is extra small its structural information can get completely lost in this process. Instead of using these degraded bottleneck features, our model injects controlled noise and refines them under physics prior guidance, enabling the network to reconstruct the missing targets before the decoder turns them into segmentation map. PhyDiff U-Net replaces multi step diffusion sampling with a single-step feature perturbation and refinement mechanism at the network bottleneck to enhance small targets. We can see the ablation results as the evidence of the effectiveness. Application of Gaussian perturbation during training and physics-guided refinement, weakened targets are restored as well as a single forward pass maintains

inference efficiency. When FRE is removed, IoU in NUAA-SIRST decreases from 0.7687 to 0.6950. The value of nIoU also decreases from 0.7379 to 0.6680. On NUDT-SIRST, IoU decreases from 0.9120 to 0.9000, proving the benefits of bottleneck refinement in real world background variability.

5.10. Hallucination Risk

There is a risk of hallucinating false targets in generative models because structured clutters like terrain edges, cloud boundaries can mimic infrared small targets. This risk has been reduced in our proposed model, it does not generate an unconstrained image, rather restricts the noise-perturbation-guided feature refinement operation to the feature domain. It does not simply replace the original bottleneck representation, instead the FRE module only applies a small residual correction scaled by a learnable parameter initialized to 0.2. This way the original encoded features are preserved. In our achieved results with PhyDiff U-Net we can observe low false alarm rate on IRSTS-1k an F_a of 16.63 while simultaneously achieving a P_d of 0.92. The results demonstrate that the feature refinement mechanism does not only construct structures mimicking the target indiscriminately.

5.11. Frequency Enhancement

Infrared small targets occupy an extremely limited spatial extent, so their structural information easily get lost during standard network downsampling and feature extraction. This drawback is addressed in our proposed architecture in multiple stages like DWT based decomposition, frequency prior extraction and wavelet based reconstruction. The encoder receives additional information from the frequency component of PPE on localized spatial variations. Even if the target is not the most highlighted region in the image, the informative changes in that region are highlighted in frequency representation. It is different from thermal prior that focuses on local intensity contrast. Discrete Wavelet Transform (DWT) frequency representation isolates rapid spatial variations, letting the model capture less highlighted targets than the background clutter. The details are not overlooked during downsampling, instead DWT separates low-frequency data from horizontal, vertical, and diagonal high-frequency details (LH, HL, HH sub-bands) contributing to target localization and boundary recovery. These preserved directional components are reconstructed by the decoder utilizing the Inverse Discrete Wavelet Transform (IWT). It prevents the ultra small masks from getting degraded. In the pre processing framework target regions are explicitly amplified, at the same time smooth and uninformative background is systematically suppressed.

5.12. Multi-Scale Aggregation and Boundary Recovery

There is significant variation in sizes of infrared targets ranging from 1 to 30 pixels. As spatial details get lost in the process, standard architectures struggle regarding over segmentation and accurately locating a target but incorrectly bleeding its predicted mask into the surrounding clutter. Our proposed decoder addresses this requirement through the Inverse Discrete Wavelet Transform (IWT) based upsampling and encoder decoder skip connections. Thus spatial maps using both low and high frequency directional sub-bands preserved in downsampling are dynamically reconstructed. Background noise that is not required is suppressed in the encoder features, after that they are fused into the skip connections. Spatial attention and the Sobel Boundary Attention (SDA) module enhance boundary recovery even further. SDA calculates horizontal and vertical gradient responses and maps local intensity transitions. This creates a soft boundary attention mask. The mask binds target boundaries and also restricts over segmentation. In the target size analysis we witness our proposed model achieving an IoU of 98.83% and a P_d of 99.68% on the target subset of 1-3 pixels. This shows an impactful implementation of the model on extremely small targets.

5.13. Practical Value of Uncertainty Estimation

All predictions are treated with equal confidence in deterministic models. The same binary weight is assigned to both strong targets and ambiguous clutter which leads to false alarms. The predicted log variance architecture incorporates an uncertainty estimation head resulting in a spatial uncertainty map through a heteroscedastic formulation. The uncertainty aware loss function enables the network to dynamically assign higher uncertainty metrics to visually ambiguous regions like target boundaries or dim objects and not to uniform background pixels. Responses with lower confidence can be suppressed instead of accepting every positive pixel with a probability value greater than 0.5. It gives a mechanism that would balance target segmentation sensitivity against strict false-alarm rejection. On IRSTS-1K, PhyDiff U-Net achieves a P_d of 0.9283 while maintaining an F_a of only 0.0043. The result shows that a large amount of background responses are not required to get accepted for the correctness of target localization.

5.14. Limitations

The proposed model achieves competitive or state-of-the-art performance on the NUAA-SIRST, IRSTS-1k and the NUDT-SIRST benchmarks, especially for probability of detection (P_d) and false-alarm rate (F_a). But, the method still has some notable limitations to be addressed.

Noise-perturbation-guided feature inference introduces additional computational cost. The noise-perturbation-guided feature bottleneck enhancement, the three auxiliary deep-supervision heads, and the multi-scale physics-prior extraction branches together add more parameters and computation over a minimal U-Net baseline as seen in Table 6. Our model uses 3.16 M parameters, and 6.605 ms latency per frame at 151.39 FPS, versus 0.40 M parameters and 3.041 ms latency at 328.84 FPS for the lighter ACM-UNet or the ALCNet with 0.427 M parameters and 3.957 ms latency per frame at 252.72 FPS. The method remains substantially more efficient than top-performing alternatives with 16× fewer parameters and 6.4× higher frame rate than UIU-Net, which requires 50.54 M parameters and 25.30 ms per frame at 23.47 FPS in comparison. Thus the diffusion and auxiliary components do introduce a measurable overhead that may call for compression in the areas of quantization, pruning, distillation before deployment on the lowest-power infrared seeker hardware.

Noise restoration may potentially create unsupported target-like structures. As reviewed in Section 2.3, unconstrained conditional diffusion frameworks are vulnerable to target hallucination. Our PhyDiff U-Net mitigates this through physics-prior-conditioned single-pass residual enhancement rather than using a multi-step unconstrained denoising loop. Empirically, the method can be seen recording the lowest or near-lowest false-alarm rates on all three benchmarks ($F_a = 3.34$ on NUAA-SIRST, 16.63 on IRSTS-1k, and 12.93 on NUDT-SIRST, as seen in Table 2), with false positives being concentrated exclusively on real clutter edges rather than invented structure. Nevertheless, the generative residual formulation does carry a residual theoretical risk. If ever put under extreme out-of-distribution scenes which were not present in the training corpus, an example, active infrared countermeasure flares, laser dazzle events, or fully saturated detector frames, the restored features could in principle produce activations resembling the true infrared targets. This possibility cannot be exhaustively ruled out by three finite benchmarks alone.

The thermal/PSF priors may require a sensor-dependent parameter adaptation. This is specifically due to the physics-prior extraction module using hand-set default values for the Gaussian point-spread-function width and the annulus local-contrast radii, where all four pyramid-level instantiations share these defaults without per-sensor override. These values are optimized for combination of statistics over the three datasets and still reach the state-of-the-art overall performance, as can be seen by the contrast with physics-blind alternatives (DNANets false-alarm rate is 9.75 on NUAA-SIRST versus 3.34 for the proposed method, see Table 2). However, they are not individually optimized per sensor modality. If a future extended study is conducted on the per-dataset ablation for these sensor-dependent hyperparameters, it can be expected to yield measurable further improvements consistent with the broader IRSTS physics-prior literature.

Another additional notable limitation can be seen for the heteroscedastic uncertainty head which only quantifies aleatoric (data-intrinsic) uncertainty and does not capture epistemic (model-intrinsic) uncertainty arising from the weight ambiguity or out-of-distribution sensor shift. A full hybrid UQ framework combining the present head with test-time MC-dropout or a small deep ensemble are areas for future possible work, consistent with the findings of Kendall & Gal (2017) Kendall and Gal (2017).

Two additional constraints are tied to the canonical per-dataset evaluation protocol and the inherent properties of the three public datasets. The end-to-end performance is bounded by the quality, native resolution, and annotation consistency of the paired training corpora. The NUAA-SIRST dataset contains sub-pixel targets whose ground-truth mask preparation introduces mild over-dilation bias. When NUDT-SIRST frames are resampled to a fixed 256×256 input grid, this subtly reduces contrast on the smallest 2–3 pixel targets. The IRSTS-1k, which aggregates imagery from multiple sensor sources with heterogeneous annotation protocols, are seen to introduce mild label noise. So for sub-pixel localization, accurate target-size estimation, and tight-intersection-over-union scoring, these dataset-level fidelity ceilings can directly affect the model's output.

Finally, the current method focuses on single-frame imagery. Following the canonical evaluation protocol of the three public benchmarks, all are curated as single-frame corpora for reproducible comparison, the proposed architecture processes each infrared frame independently, with no temporal modeling, recurrent state, or sequence-level loading. The diagnostic temporal cues available in the original 25–60 Hz continuous capture streams from which the datasets are sourced, including the smooth near-constant-velocity motion typical of airborne threats and the transient jitter of maritime wave glint, are therefore unused. The broader IRSTS literature creates the possibility of such temporal cues which can substantially reduce false alarms in maritime and aerial sequences.

6. Conclusion

Infrared small target segmentation remains a difficult mission to accomplish because of the adversity it faces from the beginning. The target size is of a few pixels with limited structural information and it exhibits weak contrast against a drastically variant background clutter. These challenges are identified and addressed in our proposed PhyDiff U-Net model that is established with integration of complementary imaging priors that have contextual and boundary aware feature learning. Local thermal contrast and high frequency, these physics components are captured by the physics prior extraction module. They provide information that reaches far beyond conventionally learned interpretations. While training at the bottleneck, the diffusion inspired enhancement module leads to controlled Gaussian perturbation. It helps refine the corrupted batch and strengthen weak target responses with the extracted physics prior, simultaneously saving the computational cost of iterative diffusion inference. Wavelet-based DWT/IWT operations next preserve the spatial and frequency information during resolution changes. Physics-noise-perturbation-guided refinement, spatial attention, and Sobel-based boundary attention combinedly work for a better target localization, whereas heteroscedastic uncertainty head gives pixel wise information on confidence that work well in cluttered areas and ambiguous boundaries. NUAA-SIRST, IRSTS-1K, and NUDT-SIRST are the three datasets used in this model demonstrating a strong result of IoU 76.87%, 65.75%, and 91.20%, respectively. The framework functions perfectly with a minimal number of 3.16M parameters and 5.732 GFLOPs. For the future experiments, there are scopes of investigation in diffusion distillation and one or few step refinement. Adaptive PSF and sensor aware prior modeling are some areas to explore with multi frame temporal information.

Availability of data

The datasets used in this study are publicly available.

The NUAA-SIRST, IRSTS-1K, and NUDT-SIRST datasets used for evaluation are publicly available from their respective sources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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